# OPPERMANN INTERVALS IN $\mathbb{F}_q[t]$: GALOIS OBSTRUCTIONS, EXACT COUNTS FOR SMALL DEGREE, AND THE OPPERMANN SUBGROUP IN CHARACTERISTIC 2

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ABSTRACT. For a monic $f \in \mathbb{F}_q[t]$ of degree $d$, the Legendre interval $\{f^2 + s : \deg s \le d\}$ splits into the $q$ *Oppermann sub-intervals* $O_{f,c} = \{f^2 + cf + s : \deg s < d\}$, each of size $q^d$. We ask whether every $O_{f,c}$ contains an irreducible polynomial.

In characteristic 2 the answer is no, and the failures have three Galois-theoretic causes: (A) for $d = 1$ the piece $c = 0$ consists of squares; (B) for $d = 2$, Frobenius acts on the roots of an additive polynomial through $\mathrm{AGL}_2(\mathbb{F}_2) \cong S_4$, and the resulting exact counts vanish exactly for $q = 2$ and $q = 4^j$; (C) for $q = 2$, $d = 4$, the Stickelberger–Swan sign of Frobenius obstructs. The additive mechanism of (B) disappears for $d = 2^{r-1} \ge 4$, because $\mathrm{AGL}_r(\mathbb{F}_2)$ contains no $2^r$-cycle when $r \ge 3$.

For odd $q$ and $d = 2$, a ternary trace form on $\mathbb{F}_{q^4}$ gives $\pi = (q^2 - 1 - (q - \eta(-1))\eta(2e))/4 > 0$. In characteristic 2 and $d = 3$, a count in the tower $\mathbb{F}_q \subset \mathbb{F}_{q^3} \subset \mathbb{F}_{q^6}$ together with a cubic Gauss sum gives every class exactly. There are $q(q^2-1)/6$ irreducibles for $k$ odd. For $k$ even there are $q(q-1)^2/6$, $q(q^2+1-2\varepsilon\sqrt{q})/6$ or $q(q^2+1+\varepsilon\sqrt{q})/6$, according as the $t^3$-coefficient is 0, a cube or a non-cube, where $\varepsilon = (-1)^{k/2}$. So $d \le 2$ is settled for every $q$, and $d = 3$ for every even $q$.

For fixed $d \ge 3$ and large $q$, the theorem of Bank, Bary-Soroker and Rosenzweig, together with the $d = 3$ count, gives nonemptiness. Exact counts from the Hayes explicit formula, evaluated by FFT, give nonemptiness in odd characteristic for $q = 3$, $d \le 16$; $q = 5$, $d \le 10$; $q = 7$, $d \le 9$; $q = 9$, $d \le 8$; $q = 11$, $d \le 7$; $q = 13$, $d \le 6$. In characteristic 2 they find no further empty sub-intervals for $q = 2$, $d \le 26$; $q = 4$, $d \le 13$; $q = 8$, $d \le 7$; $q = 16$, $d \le 5$.

Finally, in characteristic 2 the Oppermann classes form a subgroup $H_d$ of the Hayes group: the kernel of the odd power sums, with $G_d/H_d \cong \mathbb{F}_q^{\lfloor d/2 \rfloor}$. Its annihilator yields Artin–Schreier sums. The mean of $\Psi$ over $H_d$ drifts by $-2q^2 + 2q$ at $d = 4$ with $k$ odd and by $q^2 - q$ at $d = 5$. The variance is a coherent sum over annihilator cosets, maximal at $d = 2$, where $\mathrm{Var}_{H_2}(\Psi)/q^2 = q - 1$.

## 1. INTRODUCTION

Oppermann's conjecture [8] asserts that for every $n \ge 2$ both $(n^2, n^2+n)$ and $(n^2+n, (n+1)^2)$ contain a prime. It implies Legendre's conjecture and lies beyond the Riemann hypothesis, which gives primes only in intervals $[x, x + C\sqrt{x}\log x]$. Over $\mathbb{F}_q[t]$, Legendre intervals $I_f = \{f^2 + s : \deg s \le d\}$ ($f$ monic of degree $d$) were studied in [4]. Since $f^2 + cf$ and $f^2 + ct^d$ differ by a polynomial of degree $< d$, the interval $I_f$ is the disjoint union of the $q$ sub-intervals

$$O_{f,c} = \{f^2 + cf + s : \deg s < d\} = \{f^2 + ct^d + s : \deg s < d\}, \qquad c \in \mathbb{F}_q, \tag{1}$$

each of cardinality $q^d = |f|$, the norm of $f$, exactly as $(n^2, n^2+n)$ has length $n$. For $q = 2$ there are two halves. We call $O_{f,c}$ the *Oppermann sub-intervals* and ask:

*Does every $O_{f,c}$ contain an irreducible polynomial?*

The answer depends on the characteristic.

**Theorem 1.1** (characteristic 2)**.** *Let $q = 2^k$.*

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(A) *For $d = 1$ the sub-interval $O_{f,0} = \{t^2 + s_0 : s_0 \in \mathbb{F}_q\}$, which is the same set for all $q$ polynomials $f$, consists of squares. Every $O_{f,c}$ with $c \neq 0$ contains exactly $q/2$ irreducibles.*

(B) *For $d = 2$ and $f = t^2 + a_1 t + a_0$, the number of irreducibles in $O_{f,c}$ depends only on $e = a_1^2 + c$ and equals*

| | $e = 0$ | $e \neq 0$ |
|---|---|---|
| *$k$ even* | $0$ | $q^2/4$ |
| *$k$ odd* | $q(q-1)/2$ | $q(q-2)/4$ |

*The table has two zero entries: $k$ even with $e = 0$, and $q(q-2)/4$ at $q = 2$ (where $k = 1$ is odd). Hence $O_{f,c}$ is empty if and only if either $k$ is even and $c = a_1^2$, or $q = 2$ and $c = a_1^2 + 1$. For $q = 2$ the block $e = 0$ is* not *empty: it contains $q(q-1)/2 = 1$ irreducible, $t^4 + t + 1$. For every $q \in \{2, 4, 16, 64, \dots\}$ there is thus exactly one empty sub-interval as a set: $B_0$, or $B_1$ when $q = 2$. It is represented by $q^2$ pairs $(f, c)$, one $c$ for each of the $q^2$ polynomials $f$. For $q \in \{8, 32, 128, \dots\}$ there is none.*

(C) *For $q = 2$ and $d = 4$ the block $Z_4 = \{t^8 + s : \deg s \leq 3\}$ contains no irreducible polynomial. It equals $O_{f,c}$ for the 8 pairs $f = t^4 + a_2 t^2 + a_1 t + a_0$ (that is, $a_3 = 0$), $c = a_2$.*

**Theorem 1.2** (odd characteristic, computer-assisted)**.** *Let $q$ be odd. Every Oppermann sub-interval $O_{f,c}$ contains an irreducible polynomial whenever*

$$q = 3,\ d \leq 16; \qquad q = 5,\ d \leq 10; \qquad q = 7,\ d \leq 9;$$
$$q = 9,\ d \leq 8; \qquad q = 11,\ d \leq 7; \qquad q = 13,\ d \leq 6.$$

*Equivalently (Lemma 2.1), every interval $\{P + s : \deg s < d\}$ with $P$ monic of degree $2d$ contains an irreducible polynomial in these ranges.*

**Theorem 1.3** (characteristic 2, computer-assisted)**.** *For $q = 2$, $d \leq 26$; $q = 4$, $d \leq 13$; $q = 8$, $d \leq 7$; $q = 16$, $d \leq 5$, the only empty Oppermann sub-intervals are those of Theorem 1.1.*

**Theorem 1.4** (odd characteristic, $d = 2$)**.** *Let $q$ be odd and let $\eta$ be the quadratic character of $\mathbb{F}_q$, with $\eta(0) = 0$. For $f = t^2 + a_1 t + a_0$ and $c \in \mathbb{F}_q$, put $e = 2a_0 + c - a_1^2/2$. Then*

$$\pi(O_{f,c}) = \frac{q^2 - 1 - (q - \eta(-1))\,\eta(2e)}{4}.$$

*In particular, every Oppermann sub-interval with $d = 2$ contains at least $q(q-1)/4$ irreducibles if $q \equiv 1 \pmod 4$, and at least $(q+1)(q-2)/4$ if $q \equiv 3 \pmod 4$. Together with Theorem 1.1(B), this settles $d = 2$ for every $q$.*

The additive mechanism behind (B) does not extend: for $d = 2^{r-1} \geq 4$ no affine polynomial of degree $2d$ is irreducible (Theorem 3.6, Corollary 3.7).

For fixed $d$ and large $q$, the theorem of Bank, Bary-Soroker and Rosenzweig [1] covers most classes (Corollary 6.2). The one family it misses for $d \geq 3$ lies at $d = 3$ in characteristic 2, and there every class can be counted exactly.

**Theorem 1.5** (characteristic 2, $d = 3$)**.** *Let $q = 2^k$, $\varepsilon = (-1)^{k/2}$ for $k$ even, and let $c$ be the coefficient of $t^3$ in an Oppermann sub-interval $C$ with $d = 3$. Then*

$$\pi(C) = \begin{cases} q(q^2-1)/6, & k \text{ odd}, \\ q(q-1)^2/6, & k \text{ even},\ c = 0, \\ q(q^2 + 1 - 2\varepsilon\sqrt{q})/6, & k \text{ even},\ c \in (\mathbb{F}_q^\times)^3, \\ q(q^2 + 1 + \varepsilon\sqrt{q})/6, & k \text{ even},\ c \notin (\mathbb{F}_q^\times)^3. \end{cases}$$

*In particular, no Oppermann sub-interval with $d = 3$ is empty in characteristic 2.*

**Corollary 1.6.** *For $d \in \{1, 2\}$, the answer is known exactly for every $q$ (Theorems 1.1(A),(B) and 1.4). In characteristic 2 this also holds for $d = 3$ (Theorem 1.5), and no sub-interval is empty there. For every $d \geq 3$ there is $q_0(d)$ such that every Oppermann sub-interval of degree $2d$ over $\mathbb{F}_q$ with $q \geq q_0(d)$ contains an irreducible polynomial.*

For $d \geq 3$, Corollary 6.2 covers every class except those with $q$ even, $d = 3$ and $p_3 = 0$. Theorem 1.5, proved as Theorem 5.2, covers those, and indeed every class with $q$ even and $d = 3$. In characteristic 2 and $d = 2$, the empty sub-intervals for $q = 2, 4, 16, 64, \ldots$ show that no such $q_0(2)$ exists there.

The tables above therefore matter mainly for $q < q_0(d)$; $q_0(d)$ is not explicit.

In characteristic 2 the Oppermann sub-intervals have additional structure.

**Theorem 1.7** (the Oppermann subgroup)**.** *Let $q = 2^k$, and let $H_d \subset G_d$ be the set of Oppermann classes of level $d$.*

1. *$H_d$ is the kernel of the odd power-sum homomorphism $P \mapsto (S_1, S_3, \ldots, S_{2\lfloor d/2 \rfloor - 1})$, so $G_d/H_d \cong \mathbb{F}_q^{\lfloor d/2 \rfloor}$ (Theorem 2.4). The characters of $G_d/H_d$ have Artin–Schreier sums as explicit-formula coefficients (Theorem 2.5).*
2. *$\mathbb{E}_{H_d}[\Psi] - q^d = 0$ for $d = 2, 3$, and for $d = 4$ with $k$ even. It equals $-2q^2 + 2q$ for $d = 4$ with $k$ odd, and $q^2 - q$ for $d = 5$ (Corollary 7.2, Theorem 7.3).*
3. *$\mathrm{Var}_{H_2}(\Psi)/q^2 = q - 1$, whereas $\mathrm{Var}_{G_2}(\Psi)/q^2 = 1 - 1/q$. Further, $\mathrm{Var}_{H_3}(\Psi)/q^3 = 1 - 1/q$ for $k$ odd and $3(1 - 1/q)$ for $k$ even (Theorem 7.5).*

Throughout, a *sub-interval* means the set $O_{f,c}$. Distinct pairs $(f, c)$ can give the same set; counts of sub-intervals are counts of sets, that is, of Hayes classes.

*Outline.*

- Section 2 identifies the Oppermann sub-intervals with Hayes classes. In characteristic 2 it proves the Newton–trace duality.
- Sections 3–5 contain the exact algebraic results for $d \leq 3$: the three obstructions and the $\mathrm{AGL}_r(\mathbb{F}_2)$ dead zone, the ternary trace form, and the field-tower and Gauss-sum count.
- Section 6 describes the exact computation, the square-root barrier, the rigidity of the level-2 $L$-functions, and the large-$q$ corollary.
- Section 7 studies the mean and the variance over the Oppermann subgroup.

## 2. Oppermann sub-intervals as Hayes classes and Newton–trace duality

For $P$ monic of degree $n$ write $P = t^n + p_{n-1}t^{n-1} + \cdots + p_0$. Two monic polynomials of degree $n$ lie in the same *Hayes class of level $l$* if they agree in $p_{n-1}, \ldots, p_{n-l}$. Each class is a coset $P + \{s : \deg s < n - l\}$, and the map

$$P \mapsto u^n P(1/u) \bmod u^{l+1} = 1 + p_{n-1}u + \cdots + p_{n-l}u^l$$

identifies the classes with the elements of the multiplicative group $G_l = (1 + u\mathbb{F}_q[u])/(1 + u^{l+1}\mathbb{F}_q[u])$ of order $q^l$. This map is multiplicative: $\mathrm{class}(PQ) = \mathrm{class}(P)\,\mathrm{class}(Q)$ in $G_l$. The characters of $G_l$ are the short-interval characters of Hayes [5] and Keating–Rudnick [6]. A character has *exact level $j$* if it factors through $G_j$ but not through $G_{j-1}$. There are $q^j - q^{j-1}$ characters of exact level $j$, for $1 \leq j \leq l$. By (1), $O_{f,c}$ is the Hayes class of level $d$ and degree $n = 2d$ containing $f^2 + ct^d$.

**Lemma 2.1** (odd $q$)**.** *If $q$ is odd, every Hayes class of level $d$ in degree $2d$ is an Oppermann sub-interval, and it equals $O_{f,c}$ for exactly $q$ pairs $(f, c)$.*

*Proof.* Write $f = \sum_{i=0}^{d} a_i t^i$ with $a_d = 1$. For $1 \leq j \leq d$ the coefficient of $t^{2d-j}$ in $f^2$ is

$$[f^2]_{2d-j} = \sum_{\substack{i+i'=2d-j \\ 0 \leq i,i' \leq d}} a_i a_{i'} = 2a_{d-j} + \Phi_j, \qquad \Phi_j = \sum_{\substack{i+i'=2d-j \\ d-j<i,\,i'<d}} a_i a_{i'},$$

where the term $2a_{d-j} = 2a_d a_{d-j}$ comes from $\{i, i'\} = \{d, d-j\}$. The sum $\Phi_j$ runs over ordered pairs, so it includes the square $a_{d-j/2}^2$ when $j$ is even. $\Phi_j$ depends only on $a_{d-1}, \ldots, a_{d-j+1}$; for example $\Phi_1 = 0$, $\Phi_2 = a_{d-1}^2$, $\Phi_3 = 2a_{d-1}a_{d-2}$.

The map $(a_{d-1}, \ldots, a_1) \mapsto ([f^2]_{2d-1}, \ldots, [f^2]_{d+1})$ is therefore triangular with diagonal entries 2. Since $2 \in \mathbb{F}_q^\times$, it is a bijection $\mathbb{F}_q^{d-1} \to \mathbb{F}_q^{d-1}$: the prefix of a class determines $a_{d-1}, \ldots, a_1$ uniquely, by solving $a_{d-j} = ([f^2]_{2d-j} - \Phi_j)/2$ for $j = 1, \ldots, d-1$ in turn.

The remaining coefficient of the class is that of $t^d$ in $f^2 + ct^d$, namely $2a_0 + \Phi_d + c$, where $\Phi_d$ is already fixed. For each of the $q$ values of $a_0$ there is exactly one $c$ giving the prescribed value. So every class arises, and from exactly $q$ of the $q^{d+1}$ pairs $(f, c)$. □

So in odd characteristic the Oppermann question at degree $2d$ is the question whether *every* short interval of length $q^d = q^{n/2}$ contains a prime: no class is singled out.

**Lemma 2.2** ($q$ even)**.** *Let $q$ be even. The Oppermann sub-intervals are exactly the Hayes classes of level $d$ (in degree $2d$) in which the coefficient of $t^j$ is $0$ for every odd $j$ with $d < j < 2d$. The coefficients of $t^d$ and of $t^j$ for even $j$ with $d < j < 2d$ are arbitrary. There are $q^{\lceil d/2 \rceil}$ such classes among the $q^d$ classes of level $d$, and each of them equals $O_{f,c}$ for exactly $q^{\lfloor d/2 \rfloor + 1}$ pairs $(f, c)$.*

*Proof.* In characteristic 2, $f^2 = \sum a_i^2 t^{2i}$ has no odd terms, $a \mapsto a^2$ is bijective on $\mathbb{F}_q$, and the coefficient of $t^d$ is shifted by $c$. This identifies the classes that occur.

For the count, note that $a \mapsto a^2$ is additive. So the map $(f, c) \mapsto \mathrm{prefix}(f^2 + ct^d)$ is a surjective homomorphism of additive groups from $\mathbb{F}_q^{d+1}$, the coefficients $(a_0, \dots, a_{d-1}, c)$, onto $\mathbb{F}_q^{\lceil d/2 \rceil}$. Its fibres therefore all have size $q^{d+1-\lceil d/2 \rceil} = q^{\lfloor d/2 \rfloor + 1}$. □

For example, the empty classes of Theorem 1.1 at $d = 1$, 2 and 4 are represented by $q$, $q^2$ and $2^3 = 8$ pairs $(f, c)$ respectively.

Thus in characteristic 2 the Oppermann sub-intervals are the short intervals around the $q^{\lceil d/2 \rceil}$ "near-squares" $f^2 + ct^d$, a set of measure $q^{-\lfloor d/2 \rfloor}$ among all short intervals. Squares and near-squares carry Galois-theoretic structure, and that structure produces the counterexamples below.

2.1. **Newton–trace duality in characteristic 2.** Throughout this subsection $q = 2^k$, $K = \mathbb{F}_q$ and $\mathrm{Tr} = \mathrm{Tr}_{K/\mathbb{F}_2}$. We write $H_d \subset G_d$ for the set of Oppermann classes of level $d$, described in Lemma 2.2.

For monic $P = \prod_\alpha (t - \alpha)$ of degree $n$, put $\sigma(P) = u^n P(1/u) = \prod_\alpha (1 + \alpha u)$; in characteristic 2 the signs disappear. Write $\sigma(P) \equiv 1 + e_1 u + \cdots + e_d u^d \pmod{u^{d+1}}$, so that $e_j$ is the $j$-th elementary symmetric function of the roots. The class of $P$ is $\sigma(P)$ in $G_d = (1 + u\mathbb{F}_q[u])/(1 + u^{d+1}\mathbb{F}_q[u])$, a group of order $q^d$, and $\sigma(PQ) = \sigma(P)\sigma(Q)$. Let $S_m(P) = \sum_\alpha \alpha^m \in \mathbb{F}_q$. Since

$$u\frac{\sigma'(P)}{\sigma(P)} = \sum_\alpha \frac{\alpha u}{1 + \alpha u} = \sum_{m \geq 1} S_m(P) u^m$$

in characteristic 2, $S_m$ for $m \leq d$ depends only on the class of $P$, and $S_m(PQ) = S_m(P) + S_m(Q)$. Moreover $S_{2m} = S_m^2$.

**Lemma 2.3** (Newton's identities in characteristic 2)**.** *For $1 \leq j \leq d$, $S_j = \sum_{i=1}^{j-1} e_i S_{j-i} + j e_j$. In particular, for odd $j$, $e_j = S_j + \sum_{i=1}^{j-1} e_i S_{j-i}$.*

*Proof.* This is the usual identity with all signs equal to $+1$. □

**Theorem 2.4.** *Let $H_d = \{1 + \sum e_j u^j \in G_d : e_j = 0 \text{ for all odd } j < d\}$, so $|H_d| = q^{\lceil d/2 \rceil}$.*

1. *For every class, $e_j = 0$ for all odd $j < d$ if and only if $S_j = 0$ for all odd $j < d$.*
2. *The map*

$$\Phi : G_d \to (\mathbb{F}_q^{\lfloor d/2 \rfloor}, +), \qquad P \mapsto (S_1(P), S_3(P), \dots, S_{2\lfloor d/2 \rfloor - 1}(P))$$

*is a surjective homomorphism with kernel $H_d$. Hence $H_d$ is a subgroup and $G_d/H_d \cong \mathbb{F}_q^{\lfloor d/2 \rfloor}$, an elementary abelian 2-group.*

*Proof.* (1) We argue by induction over odd $j < d$. Assume both vanishing conditions hold for all odd $i < j$. In the sum $\sum_{i<j} e_i S_{j-i}$ of Lemma 2.3, every term vanishes. If $i$ is odd then $e_i = 0$. If $i$ is even then $j - i$ is odd and less than $j$, so $S_{j-i} = 0$. Hence $e_j = S_j$, which gives the inductive step in both directions.

(2) $\Phi$ is a homomorphism because each $S_m$ is additive. By (1), its kernel is $H_d$. Comparing orders, $|G_d/H_d| = q^{d - \lceil d/2 \rceil} = q^{\lfloor d/2 \rfloor}$, so $\Phi$ is surjective. □

The annihilator $H_d^\perp = \{\chi \in \widehat{G_d} : \chi|_{H_d} = 1\}$ is the dual of $G_d/H_d$. For $\beta = (\beta_1, \beta_3, \dots) \in \mathbb{F}_q^{\lfloor d/2 \rfloor}$ put

$$\chi_\beta(P) = (-1)^{\mathrm{Tr}(\sum_{j \text{ odd}<d} \beta_j S_j(P))}, \qquad g_\beta(x) = \sum_{j \text{ odd}<d} \beta_j x^j.$$

**Theorem 2.5.** *$H_d^\perp = \{\chi_\beta\}$, a group of $q^{\lfloor d/2 \rfloor}$ real characters. For every $m \geq 1$,*

$$\psi_m(\chi_\beta) := \sum_{\deg F = m} \Lambda(F)\chi_\beta(F) = \sum_{x \in \mathbb{F}_{q^m}} (-1)^{\mathrm{Tr}_{\mathbb{F}_{q^m}/\mathbb{F}_2}(g_\beta(x))}.$$

*Proof.* The first statement follows from Theorem 2.4(2). For the second, group the monic $F$ of degree $m$ with $\Lambda(F) \neq 0$ by their roots. Every $x \in \mathbb{F}_{q^m}$, of degree $s \mid m$ over $K$, determines $F_x = (\min_x)^{m/s}$, with $\Lambda(F_x) = s$; and each such $F$ arises from exactly $s$ elements $x$. By additivity, $S_j(F_x) = (m/s)\,\mathrm{Tr}_{\mathbb{F}_{q^s}/K}(x^j) = \mathrm{Tr}_{\mathbb{F}_{q^m}/K}(x^j)$. So $\chi_\beta(F_x) = (-1)^{\mathrm{Tr}_{\mathbb{F}_{q^m}/\mathbb{F}_2}(g_\beta(x))}$. The element $x = 0$ corresponds to $F = t^m$, whose class is $\sigma(t^m) = 1$; it has $\Lambda = 1$ and contributes $+1$ on both sides. No correction term is needed. □

Theorem 2.5 was checked against the exact class values for all $\beta \in \mathbb{F}_2^{\lfloor d/2 \rfloor}$, whenever $q^{2d} \leq 2^{16}$ (`verify_subgroup_mean.py`). Since $\Psi(x) = q^{-d}\sum_\chi \overline{\chi}(x)\psi_{2d}(\chi)$, the left side is $\sum_x \Psi(x)\chi_\beta(x)$.

## 3. Characteristic 2: the three obstructions and the $\mathrm{AGL}_r(\mathbb{F}_2)$ dead zone

Throughout this section $q = 2^k$ and $\mathrm{Tr} = \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}$.

3.1. **(A) $d = 1$.** Here $f = t + a_0$ and $O_{f,c} = \{t^2 + ct + s_0 : s_0 \in \mathbb{F}_q\}$.

For $c = 0$ the members $t^2 + s_0 = (t + \sqrt{s_0})^2$ are squares.

For $c \neq 0$, substitute $t = cy$: $t^2 + ct + s_0 = c^2(y^2 + y + s_0/c^2)$. This is irreducible if and only if $y^2 + y + s_0/c^2$ has no root in $\mathbb{F}_q$, that is, if and only if $\mathrm{Tr}(s_0/c^2) = 1$. Exactly $q/2$ values of $s_0$ satisfy this.

3.2. **(B) $d = 2$: an additive polynomial and $\mathrm{AGL}_2(\mathbb{F}_2)$.** For $f = t^2 + a_1 t + a_0$ we have $f^2 = t^4 + a_1^2 t^2 + a_0^2$, so

$$O_{f,c} = B_e := \{t^4 + et^2 + \alpha t + \beta : \ \alpha, \beta \in \mathbb{F}_q\}, \qquad e = a_1^2 + c.$$

By Lemma 2.2 the blocks $B_e$, $e \in \mathbb{F}_q$, are precisely the Oppermann classes for $d = 2$.

**Proposition 3.1.** *Let $\alpha \neq 0$. Put $L(t) = t^4 + et^2 + \alpha t$ and $R(y) = y^3 + ey + \alpha$.*

1. *$t^4 + et^2 + \alpha t + \beta$ is irreducible for some $\beta$ if and only if $R$ has exactly one root in $\mathbb{F}_q$.*
2. *If $R$ has exactly one root in $\mathbb{F}_q$, then $t^4 + et^2 + \alpha t + \beta$ is irreducible for exactly $q/2$ values of $\beta \in \mathbb{F}_q$.*

*If $\alpha = 0$, every member $t^4 + et^2 + \beta = (t^2 + \sqrt{e}\,t + \sqrt{\beta})^2$ is a square.*

*Proof.* $L$ is $\mathbb{F}_2$-linear and separable ($L' = \alpha \neq 0$), so $V = \ker L \subset \overline{\mathbb{F}}_q$ is a 2-dimensional $\mathbb{F}_2$-space, and $V \setminus \{0\}$ is the set of roots of $L(t)/t = R(t)$. The Frobenius $\varphi(x) = x^q$ acts on $V$ by some $F \in \mathrm{GL}(V) \cong S_3$. Exactly one of three cases holds: $F = 1$ ($R$ has three roots in $\mathbb{F}_q$), $F$ of order 3 (no root), or $F$ a transvection (exactly one root).

Fix $\beta$ and a root $r$ of $P_\beta = L + \beta$. The roots of $P_\beta$ form the coset $r + V$. Since $L(\varphi(r)) = \varphi(\beta) = \beta$, we have $\varphi(r) = r + v$ with $v \in V$, and $\varphi(r + w) = r + (Fw + v)$. So Frobenius acts on the roots through the affine map $T(w) = Fw + v$ of $V \cong \mathbb{F}_2^2$, an element of $\mathrm{AGL}_2(\mathbb{F}_2) \cong S_4$. $P_\beta$ is irreducible if and only if $T$ is a 4-cycle.

- If $F = 1$, $T$ is a translation and $T^2 = 1$.
- If $F$ has order 3, $F - 1$ is invertible, so $T$ has a fixed point.
- If $F$ is a transvection, $T^2(w) = w + (F+1)v$, so $T$ is a 4-cycle if and only if $(F+1)v \neq 0$, i.e. $v \notin V^F$.

This proves the "only if" part of (1); the "if" part follows from (2), since $q/2 \geq 1$. For (2), let $F$ be a transvection, so $(F+1)V = V^F$ is a line. Replacing $r$ by $r+w$ replaces $v$ by $v+(F+1)w$, so the class $[v] \in V/V^F \cong \mathbb{F}_2$ depends only on $\beta$, and $\beta \mapsto [v]$ is additive. Its kernel consists of the $\beta$ for which some root $r' = r+w$ is fixed by $\varphi$, i.e. $\beta \in L(\mathbb{F}_q)$. The map $L : \mathbb{F}_q \to \mathbb{F}_q$ is $\mathbb{F}_2$-linear with kernel $V^F = V \cap \mathbb{F}_q$ of order 2, so $|L(\mathbb{F}_q)| = q/2$ and exactly $q/2$ values of $\beta$ have $[v] \neq 0$. □

**Lemma 3.2.** *Let $\alpha \neq 0$. The cubic $R(y) = y^3 + ey + \alpha$ has exactly one root in $\mathbb{F}_q$ if and only if $\mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)$.*

*Proof.* The roots $y_1, y_2, y_3$ of $R$ are distinct ($R$ is separable as above), with $y_1+y_2+y_3 = 0$ and $y_1y_2y_3 = \alpha$. Put $\rho = y_1/y_3 + y_2/y_1 + y_3/y_2$ and $\rho' = y_2/y_3 + y_1/y_2 + y_3/y_1$. Even permutations of the roots fix $\rho$ and odd ones exchange $\rho, \rho'$; using $y_i + y_j = y_k$,

$$\rho + \rho' = \frac{y_1+y_2}{y_3} + \frac{y_2+y_3}{y_1} + \frac{y_3+y_1}{y_2} = 1.$$

Hence $\rho, \rho'$ are the roots of $z^2 + z + \delta$ with $\delta = \rho\rho' \in \mathbb{F}_q$, and Frobenius acts on the roots of $R$ by an odd permutation (a transposition: exactly one root in $\mathbb{F}_q$) if and only if $\rho \notin \mathbb{F}_q$, i.e. $\mathrm{Tr}(\delta) = 1$. Expanding,

$$\delta = 3 + \sum_k \frac{y_iy_j}{y_k^2} + \sum_k \frac{y_k^2}{y_iy_j} = 1 + \sum_k \frac{\alpha}{y_k^3} + \frac{1}{\alpha}\sum_k y_k^3.$$

From $y_k^3 = ey_k + \alpha$ we get $\sum_k y_k^3 = \alpha$, so $\delta = \sum_k \alpha/w_k$ with $w_k = y_k^3 = ey_k + \alpha$. Substituting $y = (w+\alpha)/e$ (for $e \neq 0$) into $w = y^3$ shows that the $w_k$ are the roots of $w^3 + \alpha w^2 + (\alpha^2+e^3)w + \alpha^3$, so $\sum_k 1/w_k = (\alpha^2+e^3)/\alpha^3$ and

$$\delta = 1 + e^3/\alpha^2.$$

For $e = 0$ this also holds directly ($w_k = \alpha$). So $\mathrm{Tr}(\delta) = 1$ if and only if $\mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)$. □

*Remark* 3.3. $\delta$ is Berlekamp's characteristic-2 discriminant [2] of $R$, and $R$ is the cubic resolvent of the quartic; the proof above is self-contained. For $e = 0$ the lemma says: $y^3 = \alpha$ has exactly one root iff $k$ is odd, which is the familiar fact that cubing is bijective on $\mathbb{F}_q$ iff $3 \nmid q-1$.

*Proof of Theorem 1.1(B).* By Proposition 3.1, $|B_e \cap \{\text{irreducibles}\}| = \frac{q}{2} \cdot \#\{\alpha \neq 0 : \mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)\}$. If $e = 0$ the condition reads $\mathrm{Tr}(1) = 1$, i.e. $k$ odd: the count is $\frac{q}{2}(q-1)$ for $k$ odd and 0 for $k$ even. If $e \neq 0$, $\alpha \mapsto e^3/\alpha^2$ permutes $\mathbb{F}_q^\times$, and $\#\{x \neq 0 : \mathrm{Tr}(x) = 1\} = q/2$, $\#\{x \neq 0 : \mathrm{Tr}(x) = 0\} = q/2 - 1$; with $\mathrm{Tr}(1) = k \bmod 2$ this gives $q^2/4$ ($k$ even) and $\frac{q}{2}(\frac{q}{2}-1)$ ($k$ odd). The count $q(q-2)/4$ vanishes only for $q = 2$. □

The formula was checked by brute force for $q \leq 64$ (script `char2_d2_theorem.py`): criterion of Lemma 3.2, the $q/2$ count of Proposition 3.1(2), and the four block totals.

3.3. **(C) $q = 2$, $d = 4$: the sign of Frobenius.** The Oppermann classes for $q = 2$, $d = 4$ are the four prefixes $(p_7, p_6, p_5, p_4) = (0, x, 0, y)$ (Lemma 2.2). For $x = y = 0$ the class is the *zero block*

$$Z_4 = \{t^8 + a_3t^3 + a_2t^2 + a_1t + a_0\} \subset \mathbb{F}_2[t],$$

equal to $O_{f,c}$ for $f = t^4 + a_2t^2 + a_1t + a_0$ and $c = a_2$. Here $a_3 = 0$ is forced, because the coefficient of $t^6$ in $f^2$ is $a_3^2$; this is why there are $2^3 = 8$ pairs rather than 16.

We use the theorem of Stickelberger [9] in the form of Swan [11, Cor. 1]: *if $\tilde{P} \in \mathbb{Z}[t]$ is monic of degree $n$ and its reduction $P \in \mathbb{F}_2[t]$ is squarefree with $r$ irreducible factors, then $r \equiv n \pmod 2$ if and only if $\mathrm{disc}(\tilde{P}) \equiv 1 \pmod 8$.* Equivalently, the Frobenius permutation of the roots of $P$ is even if and only if $\mathrm{disc}(\tilde{P}) \equiv 1 \pmod 8$. An irreducible $P$ of degree 8 has Frobenius an 8-cycle, an odd permutation, and would need $\mathrm{disc}(\tilde{P}) \equiv 5 \pmod 8$.

**Lemma 3.4.** *Let $\tilde{P} = t^n + \tilde{g}$ with $\tilde{g} \in \mathbb{Z}[t]$, $\deg \tilde{g} < n$, and $8 \mid n$. Then $\mathrm{disc}(\tilde{P}) \equiv \mathrm{Res}(\tilde{P}, \tilde{g}') \pmod 8$.*

*Proof.* $\operatorname{disc}(\tilde{P}) = (-1)^{n(n-1)/2} \operatorname{Res}(\tilde{P}, \tilde{P}') = \prod_{\tilde{P}(\rho)=0} (n\rho^{n-1} + \tilde{g}'(\rho))$, the sign being $+1$ for $8 \mid n$. Expanding the product, every term containing a factor $n$ is $n$ times a symmetric polynomial with integer coefficients in the roots, hence lies in $8\mathbb{Z}$. □

*Proof of Theorem 1.1(C).* An irreducible member of $Z_4$ has no root in $\mathbb{F}_2$, so $a_0 = 1$ and $a_1 + a_2 + a_3 = 1$. This leaves four polynomials. $t^8 + t^2 + 1 = (t^4 + t + 1)^2$ is a square. For the other three choose integer lifts and apply Lemma 3.4:

- $t^8 + t + 1$: $\tilde{g}' = 1$, $\operatorname{disc} \equiv 1$;
- $t^8 + t^3 + 1$: $\tilde{g}' = 3t^2$, $\operatorname{Res}(\tilde{P}, 3t^2) = 3^8 \big(\prod \rho\big)^2 = 3^8 \equiv 1$;
- $t^8 + t^3 + t^2 + t + 1$, lifted as $t^8 + (t+1)^3$: $\tilde{g}' = 3(t+1)^2$, and $\operatorname{Res}(\tilde{P}, 3(t+1)^2) = 3^8 \prod (\rho + 1)^2 = 3^8 \tilde{P}(-1)^2$, with $\tilde{P}(-1) = (-1)^8 + 0^3 = 1$, so $\operatorname{disc} \equiv 3^8 \equiv 1$.

Suppose one of these three were irreducible over $\mathbb{F}_2$. Then it would be squarefree with $r = 1$, and Stickelberger–Swan would require $\operatorname{disc}(\tilde{P}) \equiv 5 \pmod 8$. That contradicts $\operatorname{disc}(\tilde{P}) \equiv 1$, so none of them is irreducible. The argument does not need squarefreeness to be checked beforehand. □

*Remark* 3.5. The mechanism is that $8 \mid \deg P$ kills the leading term of $P'$ modulo 8, so the sign of Frobenius is decided by the tail $g$ alone; with $\deg g \le 3$ the tail is too short to produce the sign $-1$ on a polynomial with no linear factor. Table 1 shows the zero blocks $Z_d = \{t^{2d} + s : \deg s < d\}$, which are Oppermann classes for every $d$. For $d = 8$ ($8 \mid 2d$ again) the tail has degree $\le 7$, half of the squarefree members already have the sign $-1$, and the block contains 10 irreducibles. The parallel with (B) is exact: in (B) the obstruction is the quotient $S_4 \to S_3$ (the resolvent cubic), in (C) the quotient $S_8 \to \{\pm 1\}$ (the sign).

TABLE 1. The zero blocks $Z_d = \{t^{2d} + s : \deg s < d\} \subset \mathbb{F}_2[t]$. Columns: the squarefree members; those with $\operatorname{disc} \equiv 5 \pmod 8$, i.e. Frobenius odd (integer lift with coefficients $0, 1$); those among them with no root in $\mathbb{F}_2$; and the irreducibles. For $d = 4$ the two members with $\operatorname{disc} \equiv 5$ are $t^8 + t^3 + t$ and $t^8 + t^3 + t^2 + 1$, which have the roots 0 and 1. Stickelberger–Swan holds on every squarefree member (script `swan_parity.py`).

| $d$ | $\lvert Z_d \rvert$ | squarefree | $\operatorname{disc} \equiv 5$ | $\operatorname{disc} \equiv 5$, no root | irreducible |
|---|---|---|---|---|---|
| 2 | 4 | 2 | 2 | 1 | 1 |
| 3 | 8 | 4 | 3 | 1 | 1 |
| 4 | 16 | 8 | 2 | **0** | **0** |
| 5 | 32 | 16 | 10 | 4 | 4 |
| 6 | 64 | 32 | 20 | 8 | 6 |
| 7 | 128 | 64 | 42 | 17 | 9 |
| 8 | 256 | 128 | 64 | 24 | 10 |
| 9 | 512 | 256 | 130 | 61 | 28 |
| 10 | 1024 | 512 | 280 | 120 | 49 |

3.4. **Additive slices for $d = 2^{r-1} \ge 4$ and $\mathrm{AGL}_r(\mathbb{F}_2)$.** For $d = 2^{r-1}$, i.e. $2d = 2^r$, consider the affine polynomials

$$P_\beta(t) = L(t) + \beta, \qquad L(t) = t^{2^r} + \sum_{i=0}^{r-1} c_i t^{2^i}, \qquad c_i, \beta \in \mathbb{F}_q.$$

Apart from $t$, every monomial has even degree. In particular no odd power of $t$ lies strictly between $t^d$ and $t^{2d}$, so every such $P_\beta$ lies in a class of $H_d$. For $r = 1, 2$ (that is, $d = 1, 2$) the Oppermann classes of §3 consist of such affine cosets. There, Frobenius acts on the roots through $\mathrm{AGL}_r(\mathbb{F}_2)$, and irreducibility amounts to a $2^r$-cycle. For $r \ge 3$ this mechanism disappears.

**Theorem 3.6.** (1) *Every element $T \in \mathrm{AGL}_r(\mathbb{F}_2)$ of 2-power order satisfies $T^{2^m} = \mathrm{id}$, where $m = \lceil \log_2(r+1) \rceil$.*

(2) $\mathrm{AGL}_r(\mathbb{F}_2)$ *contains a* $2^r$*-cycle on* $\mathbb{F}_2^r$ *if and only if* $r \in \{1, 2\}$.

*Proof.* (1) Write $T(w) = Fw + v$. Since $T$ has 2-power order, so does $F \in \mathrm{GL}_r(\mathbb{F}_2)$. Hence $F^{2^s} = I$ for some $s$, so $(F+I)^{2^s} = 0$, and $N = F+I$ is nilpotent. By Cayley–Hamilton, $N^r = 0$. In characteristic 2,

$$F^{2^m} = (I+N)^{2^m} = I + N^{2^m}, \qquad I + F + \cdots + F^{2^m-1} = \frac{F^{2^m}+I}{F+I} = N^{2^m-1},$$

the second identity holding as polynomials in $F$. Therefore

$$T^{2^m}(w) = w + N^{2^m} w + N^{2^m-1} v.$$

If $2^m - 1 \geq r$, both powers of $N$ vanish and $T^{2^m} = \mathrm{id}$.

(2) A $2^r$-cycle has order $2^r$. If $r \geq 3$, then $2^{r-1} - 1 \geq r$, so the computation in (1), applied with $m = r-1$, shows that every element of 2-power order satisfies $T^{2^{r-1}} = \mathrm{id}$, a contradiction. Conversely, for $r = 1$ a translation is a 2-cycle, and for $r = 2$ a transvection with a suitable translation part is a 4-cycle Proposition 3.1. □

**Corollary 3.7** (additive dead zone)**.** *Let* $r \geq 3$. *No affine polynomial* $L(t) + \beta$ *of degree* $2^r$ *over* $\mathbb{F}_{2^k}$ *is irreducible, for any* $k$. *In particular, for* $d = 2^{r-1} \geq 4$, *the affine members of the Oppermann classes in* $H_d$ *are all reducible, and every irreducible polynomial in* $H_d$ *has a non-additive shape.*

*Proof.* If $c_0 = 0$, then $L(t) + \beta$ is a square, $\big(t^{2^{r-1}} + \sum_{i \geq 1} \sqrt{c_i}\, t^{2^{i-1}} + \sqrt{\beta}\big)^2$.

If $c_0 \neq 0$, then $L$ is separable, and its roots form an $r$-dimensional $\mathbb{F}_2$-space $V$. The roots of $L + \beta$ form a coset $\rho + V$. Since $L(\varphi(\rho)) = \beta$, we have $\varphi(\rho) = \rho + v$ with $v \in V$, so Frobenius $\varphi$ acts on the coset by $w \mapsto Fw + v$, an element of $\mathrm{AGL}_r(\mathbb{F}_2)$ (as in Proposition 3.1). Irreducibility would require a $2^r$-cycle, which Theorem 3.6 excludes. □

Both statements were checked by brute force (`verify_additive_slices.py`).

- $\mathrm{AGL}_3(\mathbb{F}_2)$ (order 1344) and $\mathrm{AGL}_4(\mathbb{F}_2)$ (order 322560) contain no $2^r$-cycle. Their largest 2-power orders are 4 and 8, so the bound $2^{\lceil \log_2(r+1) \rceil}$ is attained.
- Among the affine polynomials of degree 8 over $\mathbb{F}_q$, $q \leq 16$, and of degree 16 over $\mathbb{F}_2$ and $\mathbb{F}_4$, none is irreducible: for instance 0 of 65536 for degree 8 over $\mathbb{F}_{16}$. In degrees 2 and 4 the irreducible ones make up roughly half and a quarter, as Theorem 1.1(B) predicts.

Hence the exact AGL-counts behind Theorem 1.1(B) have no analogue at $d = 4, 8, \ldots$. There, the Oppermann classes are nonempty (Theorem 1.3; Table 6) only because of non-additive members.

## 4. Odd characteristic, $d = 2$: a ternary trace form

Throughout this section $q$ is odd and $\eta$ is the quadratic character of $\mathbb{F}_q$, with $\eta(0) = 0$.

4.1. **Reduction to one-parameter blocks.** The level-2 classes in degree 4 are $C_{p_3,p_2} = \{t^4 + p_3 t^3 + p_2 t^2 + \alpha t + \beta : \alpha, \beta \in \mathbb{F}_q\}$. The translation $t \mapsto t - p_3/4$ is defined because 4 is invertible (in characteristic 3 it is $t \mapsto t - p_3$). It preserves irreducibility and maps $C_{p_3,p_2}$ bijectively onto

$$B_e = \{t^4 + et^2 + \alpha t + \beta : \alpha, \beta \in \mathbb{F}_q\}, \qquad e = p_2 - \tfrac{3}{8} p_3^2.$$

For $O_{f,c}$ with $f = t^2 + a_1 t + a_0$ we have $p_3 = 2a_1$ and $p_2 = a_1^2 + 2a_0 + c$, so $e = 2a_0 + c - a_1^2/2$.

4.2. **The count.**

**Proposition 4.1.** *For every* $e \in \mathbb{F}_q$, *the number of irreducibles in* $B_e$ *is*

$$\pi(B_e) = \frac{q^2 - 1 - (q - \eta(-1))\eta(2e)}{4}.$$

*Proof.* A monic irreducible quartic $P \in B_e$ is the minimal polynomial of exactly four elements $x \in \mathbb{F}_{q^4}$ of degree 4, namely its roots. Write $\mathrm{Tr} = \mathrm{Tr}_{\mathbb{F}_{q^4}/\mathbb{F}_q}$. The coefficient conditions on $P$ translate into conditions on $x$: $p_3 = -\mathrm{Tr}(x)$, and when $\mathrm{Tr}\, x = 0$,

$$p_2 = e_2(x, x^q, x^{q^2}, x^{q^3}) = \tfrac{1}{2}((\mathrm{Tr}\, x)^2 - \mathrm{Tr}(x^2)) = -\tfrac{1}{2}\mathrm{Tr}(x^2).$$

Hence $4\pi(B_e) = N - M$, where

$$N = \#\{x \in W : Q(x) = -2e\}, \qquad W = \ker \mathrm{Tr}, \quad Q(x) = \mathrm{Tr}(x^2),$$

and $M$ counts the elements of $W$ with $Q(x) = -2e$ that have degree $< 4$ over $\mathbb{F}_q$, i.e. that lie in $\mathbb{F}_{q^2}$.

*The form $Q$.* The bilinear form $(x, y) \mapsto \mathrm{Tr}(xy)$ on $\mathbb{F}_{q^4}$ is nondegenerate. Its Gram determinant on a basis $\omega_1, \dots, \omega_4$ is $D = \det(\omega_i^{q^j})^2$. Frobenius permutes the columns of the matrix $(\omega_i^{q^j})$ by a 4-cycle, an odd permutation, so $\det(\omega_i^{q^j})$ is not fixed by Frobenius. It therefore lies outside $\mathbb{F}_q$ while its square $D$ lies in $\mathbb{F}_q$, so $D$ is a nonsquare in $\mathbb{F}_q$.

Since $\mathrm{Tr}(1 \cdot x) = \mathrm{Tr}\, x$, the line $\mathbb{F}_q$ is orthogonal to $W$, and $\mathbb{F}_{q^4} = \mathbb{F}_q \perp W$. The Gram determinant of $\mathbb{F}_q$ is $\mathrm{Tr}(1) = 4$, a square. So $Q|_W$ is a nondegenerate ternary form whose determinant $\Delta$ is a nonsquare.

*Counting $N$.* For a nondegenerate quadratic form in 3 variables over $\mathbb{F}_q$ with determinant $\Delta$, the number of solutions of $Q(x) = b$ is $q^2 + q\,\eta(-b\Delta)$ [7, Thm. 6.27]; for $b = 0$ this is $q^2$. With $b = -2e$ and $\eta(\Delta) = -1$ we get $N = q^2 - q\,\eta(2e)$, also for $e = 0$.

*Counting $M$.* An element of $\mathbb{F}_{q^2}$ has $\mathrm{Tr}\, x = 2\,\mathrm{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(x)$, so it lies in $W$ if and only if $x^q = -x$. Such $x$ form the line $\delta\mathbb{F}_q$, where $\delta^2 = \nu$ is a nonsquare of $\mathbb{F}_q$. For these $x$, $x^2 \in \mathbb{F}_q$ and $Q(x) = 2(x^2 + x^{2q}) = 4x^2$, so the condition is $x^2 = -e/2$. The squares of elements $s\delta$ are 0 and the nonsquares of $\mathbb{F}_q$, each nonsquare attained twice. Hence $M = 1$ if $e = 0$, and $M = 1 - \eta(-e/2)$ if $e \neq 0$; in both cases $M = 1 - \eta(-2e)$.

Therefore

$$4\pi(B_e) = q^2 - q\,\eta(2e) - 1 + \eta(-2e) = q^2 - 1 - (q - \eta(-1))\eta(2e). \qquad \square$$

*Proof of Theorem 1.4.* Combine the reduction with Proposition 4.1. The minimum over $e$ is attained at $\eta(2e) = 1$. It equals $(q^2 - 1 - (q-1))/4 = q(q-1)/4$ when $\eta(-1) = 1$, i.e. $q \equiv 1 \pmod 4$, and $(q^2 - 1 - (q+1))/4 = (q+1)(q-2)/4$ when $q \equiv 3 \pmod 4$. Both are positive for every odd $q$. $\square$

*Remark* 4.2. Brute-force counts confirm the formula for every block $B_e$ when $q = 3$, 5, 7, 9, 11, 13, 25, 27, 49, 81, and for every class $C_{p_3,p_2}$ when $q \leq 13$ (script `odd_d2_formula.py`). They also agree with the minima of Table 3. Characteristic 3 needs no special treatment.

The contrast with characteristic 2 is instructive. There the translation $t \mapsto t + a$ cannot remove $p_3$, since $4 = 0$. The Oppermann blocks are exactly the classes with $p_3 = 0$, on which $t^4 + et^2 + \alpha t$ is *additive*, and this additivity produces the obstruction of Proposition 3.1. In odd characteristic the trace form $\mathrm{Tr}(x^2)$ has the same shape for every $e$, and the count never drops below about $q^2/4 - q/4$.

## 5. Characteristic 2, $d = 3$: field-tower count and cubic Gauss sums

Let $q = 2^k$. By Lemma 2.2, the $q^2$ Oppermann classes for $d = 3$ are

$$C_{e,c} = \{t^6 + et^4 + ct^3 + s_2t^2 + s_1t + s_0 : \ s_2, s_1, s_0 \in \mathbb{F}_q\}, \qquad e, c \in \mathbb{F}_q.$$

Write $C_e = C_{e,0}$. Every member of $C_{e,0}$ has $P' = s_1$, so [1] does not apply to these classes (§6.4). The classes with $c \neq 0$ are covered by [1] for large $q$, but the count below makes this unnecessary.

**Proposition 5.1.** (1) *For each $c$, the number $\pi(C_{e,c})$ does not depend on $e$.*
(2) *$E(C_{e,0}) = q^2$, so $\Psi(C_{e,0}) = 6\pi(C_{e,0}) + q^2$. For $c \neq 0$, $E(C_{e,c}) = 0$.*

*Proof.* (1) In characteristic 2 we have $(t+a)^6 = (t^4+a^4)(t^2+a^2) = t^6 + a^2t^4 + a^4t^2 + a^6$ and $(t+a)^4 = t^4 + a^4$. Neither has a $t^3$ term. On the other hand, $c(t+a)^3 = ct^3 + cat^2 + ca^2t + ca^3$. So $t \mapsto t+a$ maps $C_{e,c}$ bijectively onto $C_{e+a^2,c}$ and preserves irreducibility. Since $a \mapsto a^2$ is bijective, the claim follows.

(2) We count the proper prime powers of degree 6 that lie in some $C_{e,c}$. Sixth powers of linear polynomials and squares of cubics contain only even powers of $t$. Cubes of irreducible quadratics never lie in any $C_{e,c}$, as shown below. So no such prime power has a $t^3$ term, and $E(C_{e,c}) = 0$ for $c \neq 0$. Now take $c = 0$ and, by (1), $e = 0$.

- *Sixth powers.* $(t+a)^6 \in C_0$ only for $a = 0$; this is $t^6$, with $\Lambda = 1$.
- *Cubes.* For a quadratic $\varpi = t^2 + bt + c$, the coefficient of $t^5$ in $\varpi^3$ is $b$. So $\varpi^3 \in C_0$ forces $b = 0$, but then $\varpi = (t + \sqrt{c})^2$ is not irreducible.
- *Squares.* For a cubic $\varpi = t^3 + a_2t^2 + a_1t + a_0$ we have $\varpi^2 = t^6 + a_2^2t^4 + a_1^2t^2 + a_0^2$, which lies in $C_0$ exactly when $a_2 = 0$. Translation changes $a_2$ by $3c = c$, so the $(q^3-q)/3$ irreducible cubics are equidistributed over the values of $a_2$, and $(q^2-1)/3$ of them have $a_2 = 0$. Each contributes $\Lambda = 3$.

Hence $E(C_0) = 1 + (q^2-1) = q^2$. The cubes are excluded from every $C_{e,c}$, not only from $C_0$, since all of these classes have $p_5 = 0$. □

**Theorem 5.2.** *Let $q = 2^k$ and $e, c \in \mathbb{F}_q$.*

(1) *If $k$ is odd, then $\pi(C_{e,c}) = q(q^2-1)/6$ for all $e, c$.*
(2) *If $k$ is even, then*

$$\pi(C_{e,c}) = \frac{q(q-1)^2}{6} + q\,m(c), \qquad m(c) = \#\{\nu \in \mathbb{F}_q^\times \setminus (\mathbb{F}_q^\times)^3 : \ \mathrm{Tr}(c/\nu) = 1\}.$$

*Here $m(0) = 0$, and $(q+2)/6 \leq m(c) \leq q/2$ for $c \neq 0$.*

*In particular, every Oppermann sub-interval with $d = 3$ in characteristic 2 contains at least $q(q-1)^2/6$ irreducibles if $k$ is even and $q(q^2-1)/6$ if $k$ is odd; this is positive for every $q$. For $c = 0$, $\Psi(C_e) = q^3 - (-1)^k(q^2-q)$.*

*Proof.* Let $K = \mathbb{F}_q \subset E = \mathbb{F}_{q^3} \subset L = \mathbb{F}_{q^6}$. Fix $c$. By Proposition 5.1(1), $6q\,\pi(C_{e,c})$ is the number of $x \in L$ of degree 6 over $K$ whose characteristic polynomial $t^6 + e_1t^5 + e_2t^4 + e_3t^3 + \dots$ has $e_1 = 0$ and $e_3 = c$.

*Step 1 (conditions on $x$).* With $S_m = \mathrm{Tr}_{L/K}(x^m)$, Newton's identity in characteristic 2 reads $S_3 = e_1S_2 + e_2S_1 + e_3$. So for $e_1 = S_1 = 0$ we get $e_3 = S_3$. The conditions are therefore $\mathrm{Tr}_{L/K}(x) = 0$ and $\mathrm{Tr}_{L/K}(x^3) = c$.

*Step 2 (the relative trace $u$).* Put $u = \mathrm{Tr}_{L/E}(x) = x + x^{q^3} \in E$; then $\mathrm{Tr}_{L/K}(x) = \mathrm{Tr}_{E/K}(u)$. Assume $\mathrm{Tr}_{E/K}(u) = 0$. We claim that $x$ has degree 6 if and only if $u \neq 0$.

- If $u = 0$, then $x^{q^3} = x$, so $x \in E$.
- If $x \in \mathbb{F}_{q^2}$, then $x^{q^3} = x^q$, so $u = x + x^q \in K$, and $0 = \mathrm{Tr}_{E/K}(u) = 3u = u$.

Since the proper subfields of $L$ are contained in $E$ or in $\mathbb{F}_{q^2}$, this proves the claim.

*Step 3 (the fibre over $u$).* Fix $u \in E^\times$ with $\mathrm{Tr}_{E/K}(u) = 0$. The $q^3$ elements $x$ with $\mathrm{Tr}_{L/E}(x) = u$ are $x = u\xi$ with $\xi + \xi^{q^3} = 1$. For these,

$$\mathrm{Tr}_{L/E}(\xi^3) = \xi^3 + (\xi+1)^3 = \xi^2 + \xi + 1.$$

Put $\alpha = \xi^2 + \xi$; then $\alpha^{q^3} = \alpha$, so $\alpha \in E$. As $\xi$ runs over the fibre, $\alpha$ runs 2-to-1 over $\{\alpha \in E : \mathrm{Tr}_{E/\mathbb{F}_2}(\alpha) = 1\}$: the equation $\xi^2 + \xi = \alpha$ has its roots outside $E$ exactly when $\mathrm{Tr}_{E/\mathbb{F}_2}(\alpha) = 1$, and then the two roots are $\xi$ and $\xi + 1 = \xi^{q^3}$. Finally,

$$\mathrm{Tr}_{L/K}(x^3) = \mathrm{Tr}_{E/K}(u^3(\alpha+1)).$$

*Step 4 (counting in the fibre).* The condition $\mathrm{Tr}_{L/K}(x^3) = c$ says that $\alpha$ lies in the affine $K$-hyperplane

$$H_{u,c} = \{\alpha \in E : \mathrm{Tr}_{E/K}(u^3\alpha) = \mathrm{Tr}_{E/K}(u^3) + c\}, \qquad |H_{u,c}| = q^2.$$

The $\mathbb{F}_2$-linear functionals on $E$ that vanish on the direction space of $H_{u,c}$ are $\alpha \mapsto \mathrm{Tr}_{E/\mathbb{F}_2}(\lambda u^3 \alpha)$ with $\lambda \in K$. Hence $\mathrm{Tr}_{E/\mathbb{F}_2}$ is constant on $H_{u,c}$ if and only if $u^3 \in K^\times$.

- If $u^3 \notin K^\times$, then $\mathrm{Tr}_{E/\mathbb{F}_2}(\alpha) = 1$ on exactly $q^2/2$ points of $H_{u,c}$. This gives $q^2$ elements $x$, whatever $c$ is.
- If $u^3 = \nu \in K^\times$, then on $H_{u,c}$ we have $\nu \,\mathrm{Tr}_{E/K}(\alpha) = \nu \,\mathrm{Tr}_{E/K}(1) + c = \nu + c$. So $\mathrm{Tr}_{E/\mathbb{F}_2}(\alpha) = \mathrm{Tr}_{K/\mathbb{F}_2}(1 + c/\nu)$ throughout $H_{u,c}$. This gives $2q^2$ elements $x$ if that value is 1, and none otherwise.

*Step 5 (summing over $u$).* There are $q^2 - 1$ admissible $u$.

- *$k$ odd.* Here $3 \nmid q^3 - 1$, so cubing is bijective on $E^\times$ and on $K^\times$, and $u^3 \in K^\times$ forces $u \in K^\times$. But every $u \in K^\times$ has $\mathrm{Tr}_{E/K}(u) = 3u = u \neq 0$, so it is not admissible. All $q^2 - 1$ admissible $u$ therefore contribute $q^2$, for every $c$, and $6q\,\pi = (q^2 - 1)q^2$.
- *$k$ even.* Here $3 \mid q - 1$, $3 \mid q^2 + q + 1$ and $\mathrm{Tr}_{K/\mathbb{F}_2}(1) = 0$. We sort the $u$ with $u^3 = \nu \in K^\times$ by $\nu$.
  - If $\nu \in (K^\times)^3$, its three cube roots lie in $K^\times$. They are not admissible, since $\mathrm{Tr}_{E/K}(u) = u \neq 0$.
  - If $\nu \notin (K^\times)^3$, its three cube roots lie in $E \setminus K$, because $K^\times \subset (E^\times)^3$. They have minimal polynomial $X^3 - \nu$, hence trace 0, and are admissible. There are $2(q-1)$ of them, and each contributes $2q^2$ when $\mathrm{Tr}_{K/\mathbb{F}_2}(c/\nu) = 1$ and nothing otherwise.
  - The remaining $(q^2 - 1) - 2(q - 1) = (q - 1)^2$ admissible $u$ contribute $q^2$ each.

  So $6q\,\pi = (q-1)^2 q^2 + 3 \cdot 2q^2\, m(c)$, which is the formula claimed.

  For the bounds on $m(c)$ when $c \neq 0$: as $\nu$ runs over the non-cubes, $c/\nu$ runs over $K^\times$ minus the coset $c(K^\times)^3$, which has $(q-1)/3$ elements. Since $K^\times$ contains $q/2$ elements of trace 1, we get $m(c) = q/2 - t(c)$ with $t(c) = \#\{y \in c(K^\times)^3 : \mathrm{Tr}(y) = 1\} \in [0, (q-1)/3]$.

The formula for $\Psi(C_e)$ follows from Proposition 5.1(2). □

5.1. **The cube-residue count $m(c)$.** It remains to evaluate $m(c)$ in Theorem 5.2(2).

**Theorem 5.3.** *Let $k$ be even, $\varepsilon = (-1)^{k/2}$, and $c \in K^\times$. Then*

$$m(c) = \begin{cases} \dfrac{q - \varepsilon\sqrt{q}}{3}, & c \in (K^\times)^3, \\ \dfrac{2q + \varepsilon\sqrt{q}}{6}, & c \notin (K^\times)^3. \end{cases}$$

*Proof.* As $\nu$ runs over the non-cubes, $y = c/\nu$ runs over $K^\times \setminus cC$, where $C = (K^\times)^3$. Hence $m(c) = q/2 - t(c)$ with $t(c) = \#\{y \in cC : \mathrm{Tr}\, y = 1\}$.

Let $\omega$ be a cubic character of $K^\times$ and $\psi_0(y) = (-1)^{\mathrm{Tr}\, y}$. The indicator of $cC$ is

$$\mathbf{1}_{cC}(y) = \tfrac{1}{3} \textstyle\sum_{i=0}^{2} \omega^i(y)\overline{\omega}^i(c),$$

so

$$\sum_{y \in cC} \psi_0(y) = \tfrac{1}{3}(-1 + \overline{\omega}(c)G(\omega) + \omega(c)G(\overline{\omega})), \qquad G(\omega) = \sum_{y \in K^\times} \omega(y)\psi_0(y).$$

Since $2 \equiv -1 \pmod 3$, the Gauss sum is pure [10], [3, §11.6]: $G(\omega) = G(\overline{\omega}) = -\varepsilon\sqrt{q}$. Hence

$$\sum_{y \in cC} \psi_0(y) = \tfrac{1}{3}(-1 - 2\varepsilon\sqrt{q}\,\mathrm{Re}\,\omega(c)),$$

where $2\,\mathrm{Re}\,\omega(c)$ equals 2 for a cube and $-1$ otherwise. Since $t(c) = \frac{1}{2}\big(|cC| - \sum_{y \in cC} \psi_0(y)\big)$ and $|cC| = (q-1)/3$, we get $t(c) = (q + 2\varepsilon\sqrt{q})/6$ for a cube and $(q - \varepsilon\sqrt{q})/6$ for a non-cube. Then $m(c) = q/2 - t(c)$. □

**Corollary 5.4.** *Let $q = 2^k$. For every characteristic-2 Oppermann class of degree 6:*

$$\pi(C_{e,c}) = \begin{cases} q(q^2-1)/6, & k \text{ odd}, \\ q(q-1)^2/6, & k \text{ even},\ c = 0, \\ q(q^2 + 1 - 2\varepsilon\sqrt{q})/6, & k \text{ even},\ c \in (K^\times)^3, \\ q(q^2 + 1 + \varepsilon\sqrt{q})/6, & k \text{ even},\ c \notin (K^\times)^3. \end{cases}$$

TABLE 2. Every characteristic-2 Oppermann class of degree 6: the number of irreducibles in $C_{e,c}$ (Theorem 5.2, Corollary 5.4). For $k$ odd all classes have the same count. For $q \le 256$ every class was also counted exactly with the Hayes FFT; for $q \ge 1024$ the two values of $m(c)$ were checked directly.

| $q$ | $\varepsilon$ | $c=0$ | $c$ cube | $c$ non-cube | $q^3/6$ |
|---|---|---|---|---|---|
| 2 | – | | 1 | | 1 |
| 4 | −1 | 6 | 14 | 10 | 11 |
| 8 | – | | 84 | | 85 |
| 16 | +1 | 600 | 664 | 696 | 683 |
| 32 | – | | 5 456 | | 5 461 |
| 64 | −1 | 42 336 | 43 872 | 43 616 | 43 691 |
| 128 | – | | 349 504 | | 349 525 |
| 256 | +1 | 2 774 400 | 2 794 880 | 2 796 928 | 2 796 203 |
| 1 024 | −1 | 178 607 616 | 178 968 064 | 178 951 680 | 178 956 971 |
| 4 096 | +1 | 11 447 654 400 | 11 453 159 424 | 11 453 290 496 | 11 453 246 123 |
| 16 384 | −1 | 732 918 276 096 | 733 008 453 632 | 733 007 405 056 | 733 007 751 851 |

*Moreover* $\Psi(C_{e,c}) = 6\pi(C_{e,c}) + q^2[c=0]$.

*Proof.* Insert Theorem 5.3 into Theorem 5.2. The prime-power term is Proposition 5.1(2). □

Thus at $d=3$, $k$ even, the counts fluctuate around $q^3/6$ by $\pm\varepsilon q^{3/2}/6$ and $\mp\varepsilon q^{3/2}/3$. This is a square-root fluctuation whose sign alternates with $k/2$, the sign of the Gauss sum. It matches the supersingular pattern $\gamma^6 \in q^3\{\pm1, \pm i\}$ observed for the level-3 zeros in Remark 5.5. Theorem 5.3 was checked for every $c$ when $q \le 1024$, and for $c=1$ and one non-cube up to $q = 2^{14}$ (Table 2; Figure 3, right).

*Remark* 5.5. For $c=0$ the error term $\pi(C_e) - q^3/6$ is $-q/6$ for $k$ odd and $-q^2/3 + q/6$ for $k$ even; for $c \neq 0$ and $k$ even it is of order $q^{3/2}$ (Corollary 5.4). This is far smaller than the $O(q^{5/2})$ a monodromy argument would give.

It reflects a supersingularity that we observed numerically: for $q \le 16$, every character of exact level 2 or 3 of $G_3$ has an $L$-function whose inverse zeros satisfy $\gamma^6 \in q^3\{\pm1, \pm i\}$. Figure 2 (right) shows these zeros for $q=8$ next to the odd case $q=7$. At level 2 this is a theorem: Proposition 6.1(3) gives $\gamma_\chi^4 = (-1)^k q^2$ for every character of exact level 2, which implies $\gamma_\chi^6 \in \{\pm q^3, \pm i q^3\}$. At level 3 it remains an observation. The proof above avoids these $L$-functions altogether. Summing over $e$ replaces the level-3 condition by the two trace conditions $\mathrm{Tr}_{L/K}(x) = 0$ and $\mathrm{Tr}_{L/K}(x^3) = c$, which the tower $K \subset E \subset L$ makes linear.

## 6. The exact computation, the square-root barrier, and large $q$

6.1. **Method.** Here $\Lambda$ is the von Mangoldt function: $\Lambda(F) = \deg \varpi$ if $F = \varpi^k$ for a monic irreducible $\varpi$ and some $k \ge 1$, and $\Lambda(F) = 0$ otherwise. For a class $x$ of level $l$ and degree $n$, let $\Psi(x) = \sum_{F \in x} \Lambda(F)$. The explicit formula [5, 6] reads

$$\Psi(x) = q^{n-l} + q^{-l} \sum_{\chi \neq 1} \overline{\chi}(x)\, \psi_n(\chi), \qquad u \frac{L'(u,\chi)}{L(u,\chi)} = \sum_{m \ge 1} \psi_m(\chi) u^m, \tag{2}$$

where for $\chi \neq 1$ the $L$-function $L(u,\chi) = \sum_F \chi(F) u^{\deg F}$ is a polynomial of degree $\le l-1$. Its coefficients are the Fourier transforms on $G_l$ of the class distributions of the monic polynomials of degree $< l$; $\psi_n(\chi)$ follows by the Newton recursion, and one inverse FFT on $G_l$ returns $\Psi(x)$ for all $q^l$ classes at once. This is the algorithm of [4], run here at level $l=d$ and degree $n=2d$. The number of irreducibles in $x$ is

$$\pi(x) = \frac{1}{n}\big(\Psi(x) - E(x)\big), \qquad E(x) = \sum_{k \ge 2,\ k \mid n} \ \sum_{\deg \varpi = n/k,\ \varpi^k \in x} \frac{n}{k}.$$

The proper prime powers in $E(x)$ are enumerated directly (Ben-Or test in degree $\le d$) and placed by the rule $\mathrm{class}(\varpi^k) = \mathrm{class}(\varpi)^k$ in $G_l$. Each run checks that every $\pi(x)$ is an integer and that $\sum_x (n\pi(x) + E(x)) = q^n$, and reports the largest distance of the floating-point $\Psi(x)$ from an integer (below $10^{-7}$ in all runs). As an independent check, a brute-force census that tests every polynomial of degree $2d$ for irreducibility (`verify_pointwise_oppermann.py`) agrees with the FFT counts class by class for all $q^{2d} \le 2^{22}$ (`compare_bruteforce.py`), and in minimum, mean and empty classes for all $q^{2d} \le 2^{26}$.

6.2. **Results.** Table 3 lists, for odd $q$, the minimum of $\pi(x)$ over all classes (all of which are Oppermann sub-intervals by Lemma 2.1) and its ratio to the expected value $q^d/2d$. Table 4 gives the same for the Oppermann classes in characteristic 2. The full output, all degrees, is in the file
`data/hayes_oppermann_census.txt`.

*Proof of Theorems 1.2 and 1.3.* Exact computation as described above; the minima are in Tables 3 and 4, and the empty classes in characteristic 2 occur exactly at

$$(q, d) \in \{(2,1), (2,2), (2,4), (4,1), (4,2), (8,1), (16,1), (16,2)\},$$

one class each, which are the classes of Theorem 1.1. □

The normalised minimum $\min_x \pi(x)/(q^d/2d)$ tends to 1 in every column: the worst Oppermann sub-interval is only slightly below average, far from the Weil bound discussed next.

TABLE 3. Odd $q$: irreducible counts $\pi$ over all $q^d$ Hayes classes of level $d$ in degree $2d$ (all are Oppermann sub-intervals). Last four degrees for each $q$; all smaller degrees are in the data file, with minimum $\ge 1$ throughout.

| $q$ | $d$ | classes | $\min \pi$ | mean $\pi$ | $\min \pi/(q^d/2d)$ |
|---|---|---|---|---|---|
| 3 | 13 | 1 594 323 | 60 556 | 61 320.1 | 0.9875 |
| | 14 | 4 782 969 | 169 300 | 170 820.3 | 0.9911 |
| | 15 | 14 348 907 | 476 000 | 478 296.9 | 0.9952 |
| | 16 | 43 046 721 | 1 341 030 | 1 345 210.0 | 0.9969 |
| 5 | 7 | 78 125 | 5 418 | 5 580.3 | 0.9709 |
| | 8 | 390 625 | 23 951 | 24 414.0 | 0.9810 |
| | 9 | 1 953 125 | 107 410 | 108 506.9 | 0.9899 |
| | 10 | 9 765 625 | 486 042 | 488 281.2 | 0.9954 |
| 7 | 6 | 117 649 | 9 500 | 9 804.0 | 0.9690 |
| | 7 | 823 543 | 57 981 | 58 824.4 | 0.9857 |
| | 8 | 5 764 801 | 358 343 | 360 300.0 | 0.9946 |
| | 9 | 40 353 607 | 2 235 915 | 2 241 867.0 | 0.9973 |
| 9 | 5 | 59 049 | 5 680 | 5 904.8 | 0.9619 |
| | 6 | 531 441 | 43 728 | 44 286.7 | 0.9874 |
| | 7 | 4 782 969 | 339 993 | 341 640.6 | 0.9952 |
| | 8 | 43 046 721 | 2 684 630 | 2 690 420.0 | 0.9978 |
| 11 | 4 | 14 641 | 1 735 | 1 830.0 | 0.9480 |
| | 5 | 161 051 | 15 804 | 16 105.0 | 0.9813 |
| | 6 | 1 771 561 | 146 403 | 147 630.0 | 0.9917 |
| | 7 | 19 487 171 | 1 388 036 | 1 391 940.7 | 0.9972 |
| 13 | 3 | 2 197 | 349 | 366.0 | 0.9531 |
| | 4 | 28 561 | 3 450 | 3 570.0 | 0.9664 |
| | 5 | 371 293 | 36 672 | 37 129.2 | 0.9877 |
| | 6 | 4 826 809 | 400 104 | 402 234.0 | 0.9947 |

TABLE 4. Even $q$: irreducible counts over the $q^{\lceil d/2 \rceil}$ Oppermann classes (Lemma 2.2). Last four degrees for each $q$; the only empty classes in the whole range are those of Theorem 1.1.

| $q$ | $d$ | classes | $\min \pi$ | mean $\pi$ | $\min \pi/(q^d/2d)$ |
|---|---|---|---|---|---|
| 2 | 23 | 4 096 | 181 349 | 182 329.3 | 0.9945 |
| | 24 | 4 096 | 347 822 | 349 412.9 | 0.9951 |
| | 25 | 8 192 | 669 169 | 671 025.2 | 0.9971 |
| | 26 | 8 192 | 1 287 576 | 1 290 408.9 | 0.9977 |
| 4 | 10 | 1 024 | 51 908 | 52 350.0 | 0.9901 |
| | 11 | 4 096 | 189 054 | 190 603.5 | 0.9916 |
| | 12 | 4 096 | 695 868 | 698 820.5 | 0.9954 |
| | 13 | 16 384 | 2 575 566 | 2 580 963.0 | 0.9979 |
| 8 | 4 | 64 | 440 | 490.0 | 0.8594 |
| | 5 | 512 | 3 204 | 3 276.0 | 0.9778 |
| | 6 | 512 | 21 616 | 21 840.0 | 0.9895 |
| | 7 | 4 096 | 149 148 | 149 796.0 | 0.9957 |
| 16 | 2 | 16 | **0** | 60.0 | **0** |
| | 3 | 256 | 600 | 680.0 | 0.8789 |
| | 4 | 256 | 7 616 | 8 160.0 | 0.9297 |
| | 5 | 4 096 | 103 680 | 104 856.0 | 0.9888 |

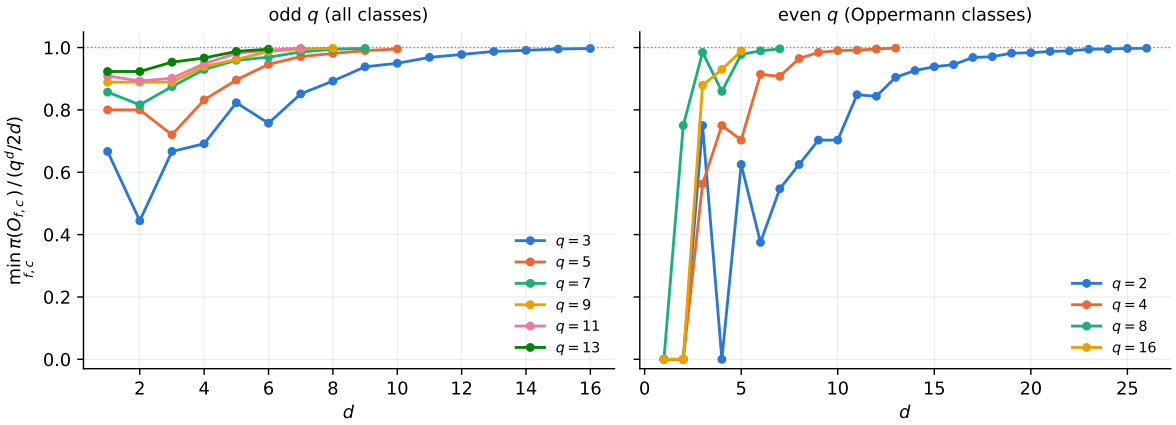



FIGURE 1. Normalised minimum $\min_{f,c} \pi(O_{f,c})/(q^d/2d)$ against $d$. Left: odd $q$. Right: even $q$; the points at 0 are the empty sub-intervals of Theorem 1.1.

6.3. **The square-root barrier.** In (2) with $n = 2d$ and $l = d$ the main term is $q^d$.

Let $\chi$ have exact level $j$. The monic polynomials of degree $\geq j$ are equidistributed over $G_j$, so $\deg L(u, \chi) \leq j - 1$. By Weil, every inverse zero $\gamma$ of $L(u, \chi)$ has $|\gamma| \in \{1, \sqrt{q}\}$, hence $|\psi_{2d}(\chi)| \leq (j-1)q^d$. Characters of level 1 have $L = 1$ and contribute nothing. Multiplying by the factor $q^{-d}$ in (2), which cancels the $q^d$, and summing over the $q^j - q^{j-1}$ characters of each exact level $j$ gives

$$|\Psi(x) - q^d| \leq \sum_{j=2}^{d} (j-1)(q^j - q^{j-1}) = (d-1)q^d - \sum_{j=1}^{d-1} q^j. \tag{3}$$

*For $d \geq 3$* the right-hand side exceeds $q^d$ for every $q$, so (3) does not even give $\Psi(x) > 0$. This is the exact analogue of the situation over $\mathbb{Z}$: Oppermann's intervals have length $\sqrt{x}$, and the

Riemann hypothesis falls short by a factor $\log x$. No degree threshold beyond which $O_{f,c} \neq \emptyset$ follows from Weil's theorem.

*For $d = 2$,* (3) reads $|\Psi(x) - q^2| \leq q^2 - q$, so $\Psi(x) \geq q > 0$ for every class. This still does not produce an irreducible, because $\Psi$ also counts prime powers. In characteristic 2 the bound is attained, and this pins down every level-2 $L$-function.

**Proposition 6.1** (characteristic 2, $d = 2$)**.** *Let $q = 2^k$, and let $B_e = \{t^4 + et^2 + \alpha t + \beta\}$ as in §3.*

1. *$E(B_e) = q$ for every $e \in \mathbb{F}_q$.*
2. *$\Psi(B_0) = q^2 - (-1)^k(q^2 - q)$, and $\Psi(B_e) = q^2 + (-1)^k q$ for $e \neq 0$. In particular, $B_0$ attains the lower Weil bound $q$ when $k$ is even and the upper bound $2q^2 - q$ when $k$ is odd.*
3. *Every character $\chi$ of exact level 2 has $L(u, \chi) = 1 - \gamma_\chi u$ with $\gamma_\chi^4 = (-1)^k q^2$.*
4. *For every level-2 class $x$ in degree 4,*

$$\Psi(x) = q^2 - (-1)^k(q^2[x = 1] - q\,[p_3(x) = 0]).$$

*In particular, every class with $p_3 \neq 0$ contains exactly $q^2/4$ irreducibles.*

*Proof.* (1) The proper prime powers of degree 4 are $\varpi^4$ with $\deg \varpi = 1$ and $\varpi^2$ with $\deg \varpi = 2$.

- $(t + c)^4 = t^4 + c^4$ lies in $B_e$ only for $e = 0$, and then for all $q$ values of $c$, each with $\Lambda = 1$.
- $(t^2 + b_1 t + b_0)^2 = t^4 + b_1^2 t^2 + b_0^2$ lies in $B_e$ exactly when $b_1^2 = e$. For $e = 0$ this forces $b_1 = 0$, and $t^2 + b_0$ is a square, not irreducible. For $e \neq 0$ we have $b_1 = \sqrt{e} \neq 0$, and by Theorem 1.1(A) the polynomial $t^2 + b_1 t + b_0$ is irreducible for exactly $q/2$ values of $b_0$, each contributing $\Lambda = 2$.

In both cases $E(B_e) = q$.

(2) We have $\Psi(B_e) = 4\pi(B_e) + q$. Insert the counts of Theorem 1.1(B).

(3) $B_0$ is the identity of $G_2$, so $\chi(B_0) = 1$ for every $\chi$. Characters of exact level 1 have $L = 1$ and $\psi_4 = 0$. By (2) and (2),

$$\sum_{\chi \text{ exact level } 2} \psi_4(\chi) = q^2(\Psi(B_0) - q^2) = -(-1)^k(q^2 - q)\,q^2.$$

There are $q^2 - q$ such characters, and each has $|\psi_4(\chi)| \leq q^2$. Equality in the triangle inequality therefore forces $\psi_4(\chi) = -(-1)^k q^2$ for each of them. Then $L(u, \chi)$ cannot be constant, so $L(u, \chi) = 1 - \gamma_\chi u$ and $\psi_4(\chi) = -\gamma_\chi^4$.

(4) By (3) and (2),

$$\Psi(x) = q^2 - (-1)^k \sum_{\chi \text{ exact level } 2} \overline{\chi}(x).$$

The inner sum is a sum over all characters of $G_2$ minus a sum over the characters of $G_1$, which equals $q^2[x = 1] - q[x \in \ker(G_2 \to G_1)]$. The kernel consists of the classes with $p_3 = 0$.

If $p_3(x) \neq 0$, then $x$ contains no proper prime power: $\varpi^4$ and $\varpi^2$ have no odd-degree terms. So $\pi(x) = \Psi(x)/4 = q^2/4$. □

All four statements were confirmed for $q = 2, 4, 8, 16$ by the script `char2_d2_rigidity.py`. In characteristic 2 the explicit formula thus degenerates completely at $d = 2$. The $q^2 - q$ characters of exact level 2 all have the same $\psi_4$, so $\Psi$ takes only three values: on the identity class, on the other classes with $p_3 = 0$, and on the classes with $p_3 \neq 0$. The Oppermann classes are exactly the classes with $p_3 = 0$ (Lemma 2.2), and the identity class $B_0$ is where the fourth powers absorb the entire Weil lower bound.

Results on the variance of $\Psi$ over Legendre intervals (an average over $f$) bound the number of intervals that deviate by a given amount; they do not exclude a single empty interval.

6.4. **Large $q$.** For fixed degree and $q \to \infty$, Bank, Bary-Soroker and Rosenzweig [1] prove the following. For monic $P \in \mathbb{F}_q[t]$ of degree $k$ and $1 \leq m < k$, the number $N$ of irreducibles in $\{P + h : \deg h \leq m\}$ satisfies

$$|N - q^{m+1}/k| \leq c(k)\,q^{m+1/2},$$

provided that $m \geq 2$ if $\gcd(q, k(k-1)) > 1$, and $m \geq 3$ if $q$ is even and $\deg P' \leq 1$. They also note that the estimate fails outside this range.

Our blocks $B_e = \{t^4 + et^2 + h : \deg h \leq 1\}$ have $k = 4$ and $m = 1$, and in characteristic 2 the centre $P = t^4 + et^2$ has $P' = 0$, so both conditions exclude them. Theorem 1.1(B) gives the exact count at this excluded point and shows how the asymptotic fails:

- for $q = 4^j$ the block $e = 0$ contains no irreducible, while the expected count is $q^2/4$;
- for $q = 2^{2j+1}$ the block $e = 0$ contains $q(q-1)/2$, about twice the expected count.

The blocks $e \neq 0$, with $q^2/4$ and $q(q-2)/4$ irreducibles, do agree with the asymptotic. The obstruction comes from the additive structure of $t^4 + et^2 + \alpha t$ (Proposition 3.1). For $m = 1$, $k = 4$ and $q$ even, the conditions of [1] therefore cannot be dropped. By contrast, when $3 \mid q$, a case that is also excluded by $\gcd(q, k(k-1)) > 1$ at $k = 4$, $m = 1$, Theorem 1.4 shows that the asymptotic $q^2/4 + O(q)$ continues to hold without exception.

Theorem 1.1(C) has $q = 2$ fixed, so it lies outside the $q \to \infty$ regime altogether.

In the other direction, [1] gives nonemptiness for large $q$ in most cases. An Oppermann sub-interval $O_{f,c} = \{f^2 + ct^d + h : \deg h \leq d-1\}$ has $k = 2d$ and $m = d-1$. Since $N \geq q^d/2d - c(2d)q^{d-1/2}$, the sub-interval is nonempty as soon as $q > (2d\,c(2d))^2$, provided the conditions of [1] hold. We do not make the constant $c(2d)$ explicit.

**Corollary 6.2** (of [1])**.** *For each $d$ there is $q_0(d)$ such that every Oppermann sub-interval of degree $2d$ over $\mathbb{F}_q$, with $q \geq q_0(d)$, contains an irreducible polynomial in each of the following cases:*

(1) *$q$ odd and $d \geq 3$;*
(2) *$q$ odd, $d = 2$ and $\gcd(q, 6) = 1$;*
(3) *$q$ even and $d \geq 4$;*
(4) *$q$ even, $d = 3$, and the class has $p_3 \neq 0$.*

*Proof.* We check the conditions of [1] with $k = 2d$ and $m = d-1$; recall that $1 \leq m < k$ is required.

- *$q$ odd.* The only condition is $m \geq 2$ when $p \mid 2d(2d-1)$. It holds for $d \geq 3$. For $d = 2$, $k(k-1) = 12$, so it holds when $\gcd(q, 6) = 1$.
- *$q$ even, $d \geq 4$.* Here $m \geq 3$, so both conditions hold.
- *$q$ even, $d = 3$.* By Lemma 2.2 every member $P$ of an Oppermann class has $p_5 = 0$. In characteristic 2 this gives $P' = p_3 t^2 + p_1$, so $\deg P' = 2$ when $p_3 \neq 0$, and the condition "$m \geq 3$ if $\deg P' \leq 1$" does not apply.

□

The cases left open by the corollary are the following. For $d = 1$ the interval is too short for [1], but Theorem 1.1(A) and, for odd $q$, the count $(q-1)/2$ of irreducible $t^2 + bt + s_0$ settle it. For $d = 2$ the corollary is superseded: Theorems 1.1(B) and 1.4 give exact counts for every $q$, including $3 \mid q$, with no threshold. For $q$ even with $d = 3$, the corollary is superseded as well. Theorem 5.2 in §5 counts all $q^2$ classes exactly for every $q$, including the $q$ classes with $p_3 = 0$ (prefixes $(p_5, p_4, p_3) = (0, x, 0)$) that [1] misses. What remains is $q < q_0(d)$ for $q$ odd with $d \geq 3$, and for $q$ even with $d \geq 4$; these ranges are covered only by computation.

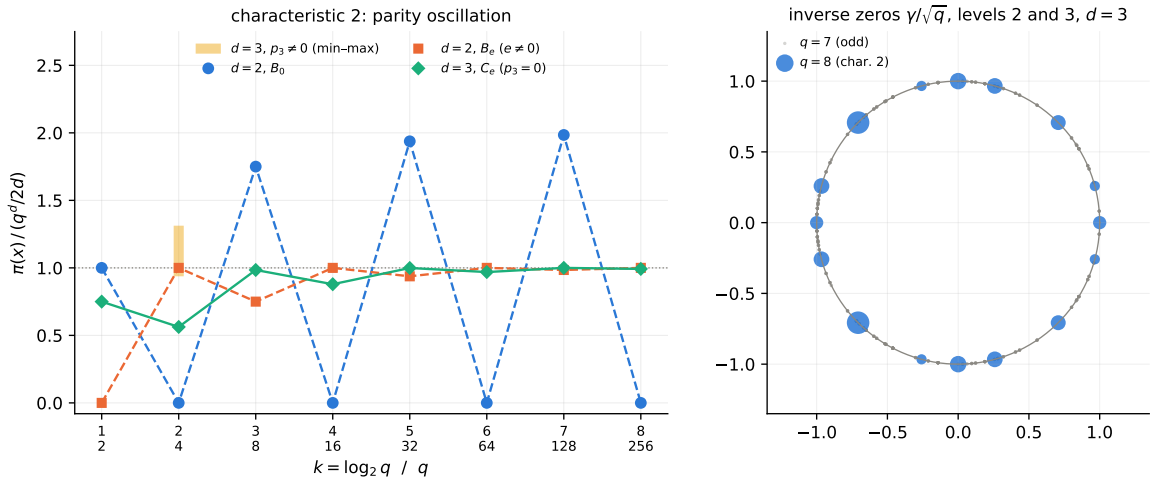



FIGURE 2. Characteristic-2 rigidity. *Left:* normalised counts $\pi(x)/(q^d/2d)$ for $q = 2^k$, $k \le 8$. Exact values from the Hayes FFT are shown as markers; the formulas of Theorem 1.1(B) and Theorem 5.2 are shown as lines. The bars give the range over the $d = 3$ Oppermann classes with $p_3 \neq 0$. For odd $k$ this range reduces to the single value of $C_e$, as Theorem 5.2(1) predicts. For even $k$ its endpoints are given by the two values of $m(c)$. *Right:* normalised inverse zeros $\gamma/\sqrt{q}$ of $L(u,\chi)$ for the characters of exact level 2 and 3 of $G_3$. For $q = 8$ (blue; marker area proportional to multiplicity) they occupy only 16 of the 24-th roots of unity. For $q = 7$ (grey; 630 zeros) they spread over the whole circle.

## 7. Mean drift and coherent variance on the Oppermann subgroup

Throughout this section $q = 2^k$, and $H_d \subset G_d$ is the Oppermann subgroup of §2.1.

### 7.1. Mean drift on $H_d$ for $d \le 5$.

**Proposition 7.1.** *Let $L = \mathbb{F}_{q^{2d}}$ and $N_d = \#\{x \in L : \mathrm{Tr}_{L/K}(x^j) = 0$ for all odd $j < d\}$. Then*

$$\mathbb{E}_{x\in H_d}[\Psi(x)] = q^{-\lceil d/2\rceil} N_d = q^d + q^{-d}\sum_{\beta\neq 0} \psi_{2d}(\chi_\beta).$$

*Proof.* $\sum_{x\in H_d}\Psi(x) = \sum \Lambda(F)$ over the $F$ of degree $2d$ with $\Phi(F) = 0$. Group these by roots as in the proof of Theorem 2.5; since $S_j(F_x) = \mathrm{Tr}_{L/K}(x^j)$, the sum equals $N_d$. The second expression follows from orthogonality: $|H_d|^{-1}\sum_{x\in H_d}\overline{\chi}(x)$ equals 1 if $\chi \in H_d^\perp$ and 0 otherwise, and $\psi_{2d}(1) = q^{2d}$. □

**Corollary 7.2.** *For $d = 2, 3$, $\mathbb{E}_{H_d}[\Psi] = q^d$. Moreover $\mathbb{E}_{H_2}[\pi] = q(q-1)/4$ and $\mathbb{E}_{H_3}[\pi] = q(q^2-1)/6$.*

*Proof.* Only $j = 1$ occurs, and $\psi_{2d}(\chi_{\beta_1}) = \sum_x (-1)^{\mathrm{Tr}(\beta_1 x)} = 0$ for $\beta_1 \neq 0$.

For $\pi$ we subtract the elements of degree less than $2d$ from $N_d$. For $d = 2$, every $x \in \mathbb{F}_{q^2}$ has $\mathrm{Tr}_{L/K}(x) = 2\,\mathrm{Tr}_{\mathbb{F}_{q^2}/K}(x) = 0$. These are $q^2$ elements, and $N_2 = q^3$, so $\sum_{H_2} 4\pi = q^3 - q^2$. For $d = 3$, the elements of $\mathbb{F}_{q^3}$ satisfy the condition automatically. An element $x \in \mathbb{F}_{q^2}$ has $\mathrm{Tr}_{L/K}(x) = \mathrm{Tr}_{\mathbb{F}_{q^2}/K}(x)$, which vanishes only for $x \in K$ (since $\ker \mathrm{Tr}_{\mathbb{F}_{q^2}/K} = K$ in characteristic 2). Hence $\sum_{H_3} 6\pi = q^5 - q^3$. □

**Theorem 7.3.** *Let $d \in \{4, 5\}$ and $q = 2^k$. Then*

| | $N_d$ | $\mathbb{E}_{H_d}[\Psi]$ | $\mathbb{E}_{H_d}[\pi]$ |
|---|---|---|---|
| $d = 4$, $k$ even | $q^6$ | $q^4$ | $(q^4 - q^2)/8$ |
| $d = 4$, $k$ odd | $q^6 - 2q^4 + 2q^3$ | $q^4 - 2q^2 + 2q$ | $(q^4 - 3q^2 + 2q)/8$ |
| $d = 5$ | $q^8 + q^5 - q^4$ | $q^5 + q^2 - q$ | $(q^5 - q)/10$ |

*Proof.* For $d \in \{4,5\}$ the odd indices are $j = 1, 3$. Let $E = \mathbb{F}_{q^d}$, so $K \subset E \subset L = \mathbb{F}_{q^{2d}}$, and put $u = \mathrm{Tr}_{L/E}(x) = x + x^{q^d}$. Then $\mathrm{Tr}_{L/K}(x) = \mathrm{Tr}_{E/K}(u)$.

*Case $u = 0$.* Then $x \in E$. Both traces vanish, since $\mathrm{Tr}_{L/K}(y) = 2\,\mathrm{Tr}_{E/K}(y) = 0$ for $y \in E$. This gives $q^d$ solutions.

*Case $u \neq 0$.* We need $\mathrm{Tr}_{E/K}(u) = 0$, which leaves $q^{d-1}-1$ values of $u$. As in Theorem 5.2, write $x = u\xi$ with $\xi + \xi^{q^d} = 1$, and put $\alpha = \xi^2 + \xi \in E$. Then $\alpha$ runs 2-to-1 over $\{\mathrm{Tr}_{E/\mathbb{F}_2}(\alpha) = 1\}$, and $\mathrm{Tr}_{L/K}(x^3) = \mathrm{Tr}_{E/K}(u^3(\alpha+1))$. The condition $\mathrm{Tr}_{L/K}(x^3) = 0$ cuts out an affine $K$-hyperplane $H_u \subset E$ of size $q^{d-1}$. On $H_u$, the functional $\mathrm{Tr}_{E/\mathbb{F}_2}$ is non-constant unless $u^3 = \nu \in K^\times$.

- If $u^3 \notin K^\times$, then $u$ contributes $q^{d-1}$ elements $x$.
- If $u^3 = \nu \in K^\times$, then $\mathrm{Tr}_{E/K}(\alpha) = \mathrm{Tr}_{E/K}(1) = d$ on $H_u$, so $\mathrm{Tr}_{E/\mathbb{F}_2}(\alpha) = dk \bmod 2$. Then $u$ contributes $2q^{d-1}$ elements if $dk$ is odd, and none otherwise.

With $a_d = \#\{u \in E^\times : u^3 \in K,\ \mathrm{Tr}_{E/K}(u) = 0\}$ this gives

$$N_d = q^d + (q^{d-1} - 1 - a_d)\, q^{d-1} + 2a_d\, q^{d-1}[dk \text{ odd}].$$

*Computing $a_d$.* Such $u$ is a root of some $X^3 - \nu$ with $\nu \in K^\times$, so its degree over $K$ is 1, 2 or 3, and it must divide $d$.

- $d = 4$. Every $u \in K^\times$ qualifies, since $\mathrm{Tr}_{E/K}(u) = 4u = 0$. Elements of degree 2 occur only when some $X^3 - \nu$ has an irreducible quadratic factor over $K$. This happens exactly when $k$ is odd: then $3 \nmid q-1$, and $X^3 - \nu = (X-a)(X^2 + aX + a^2)$. The resulting $2(q-1)$ elements $a\omega, a\omega^2 \in \mathbb{F}_{q^2} \setminus K$ have trace $\mathrm{Tr}_{E/K} = 2\,\mathrm{Tr}_{\mathbb{F}_{q^2}/K} = 0$. When $k$ is even, the cube roots of a non-cube lie in $\mathbb{F}_{q^3} \not\subset E$. So $a_4 = q-1$ for $k$ even and $a_4 = 3(q-1)$ for $k$ odd. In both cases $dk$ is even.
- $d = 5$. Only degree 1 is possible. But $u \in K^\times$ has $\mathrm{Tr}_{E/K}(u) = 5u = u \neq 0$, so $a_5 = 0$.

Substituting gives $N_d$, and $\mathbb{E}_{H_d}[\Psi] = N_d / q^{\lceil d/2 \rceil}$.

*The mean of $\pi$.* The elements of degree less than $2d$ satisfying the conditions are exactly those of $E$. For $d = 4$, every proper subfield of $\mathbb{F}_{q^8}$ lies in $E$. For $d = 5$, an element $x \in \mathbb{F}_{q^2} \setminus K$ has $\mathrm{Tr}_{L/K}(x) = 5\,\mathrm{Tr}_{\mathbb{F}_{q^2}/K}(x) \neq 0$ (since $\ker \mathrm{Tr}_{\mathbb{F}_{q^2}/K} = K$ in characteristic 2). Hence $\sum_{H_d} 2d\pi = N_d - q^d$. □

Table 5 compares Corollary 7.2 and Theorem 7.3 with the exact class counts for $q \leq 16$; all 16 cases agree. By Proposition 7.1, the drift $\mathbb{E}_{H_d}[\Psi] - q^d$ is $q^{-d}$ times a sum of Artin–Schreier sums of the Gold-type polynomials $\beta_1 x + \beta_3 x^3$ over $\mathbb{F}_{q^{2d}}$. It vanishes for $d = 4$ with $k$ even. It is $-2q^2 + 2q$ for $d = 4$ with $k$ odd, coming from the cube roots of unity outside $K$, and $+q^2 - q$ for $d = 5$. Figure 3 (left) shows the drift in units of $q^{d/2}$ for $d \leq 12$.

7.2. **Coherent superposition and subgroup variance.** By the explicit formula, $\Psi(x) = q^{-d} \sum_{\chi \in \widehat{G_d}} \overline{\chi}(x)\psi_{2d}(\chi)$. On all of $G_d$, Parseval gives

$$\mathrm{Var}_{G_d}(\Psi) = q^{-2d} \sum_{\chi \neq 1} |\psi_{2d}(\chi)|^2,$$

one term per character. On $H_d$ this changes.

**Proposition 7.4.**

$$\mathrm{Var}_{x \in H_d}(\Psi(x)) = q^{-2d} \sum_{[\chi] \neq H_d^\perp} \Big| \sum_{\eta \in H_d^\perp} \psi_{2d}(\chi\eta) \Big|^2,$$

*where $[\chi]$ runs over the cosets of $H_d^\perp$ in $\widehat{G_d}$.*

*Proof.* For $x \in H_d$, $\overline{\chi}(x)$ depends only on the restriction $\chi|_{H_d}$, that is, on the coset $[\chi] \in \widehat{G_d}/H_d^\perp \cong \widehat{H_d}$. Hence

$$\Psi|_{H_d} = q^{-d} \sum_{\lambda \in \widehat{H_d}} \overline{\lambda}\, A(\lambda), \qquad A(\lambda) = \sum_{\chi|_{H_d} = \lambda} \psi_{2d}(\chi).$$

The trivial $\lambda$ gives the mean (Proposition 7.1). Apply Parseval on $H_d$. □

TABLE 5. Means over the Oppermann subgroup $H_d$ (exact, from the Hayes FFT) against the closed forms of Corollary 7.2 and Theorem 7.3. Every entry agrees.

| $q$ | $d$ | $\lvert H_d\rvert$ | $\mathbb{E}_{H_d}[\Psi]$ | $\mathbb{E}_{H_d}[\pi]$ | $q^d$ |
|---|---|---|---|---|---|
| 2 | 2 | 2 | 4 | 1/2 | 4 |
| 2 | 3 | 4 | 8 | 1 | 8 |
| 2 | 4 | 4 | 12 | 1 | 16 |
| 2 | 5 | 8 | 34 | 3 | 32 |
| 4 | 2 | 4 | 16 | 3 | 16 |
| 4 | 3 | 16 | 64 | 10 | 64 |
| 4 | 4 | 16 | 256 | 30 | 256 |
| 4 | 5 | 64 | 1 036 | 102 | 1 024 |
| 8 | 2 | 8 | 64 | 14 | 64 |
| 8 | 3 | 64 | 512 | 84 | 512 |
| 8 | 4 | 64 | 3 984 | 490 | 4 096 |
| 8 | 5 | 512 | 32 824 | 3 276 | 32 768 |
| 16 | 2 | 16 | 256 | 60 | 256 |
| 16 | 3 | 256 | 4 096 | 680 | 4 096 |
| 16 | 4 | 256 | 65 536 | 8 160 | 65 536 |
| 16 | 5 | 4 096 | 1 048 816 | 104 856 | 1 048 576 |

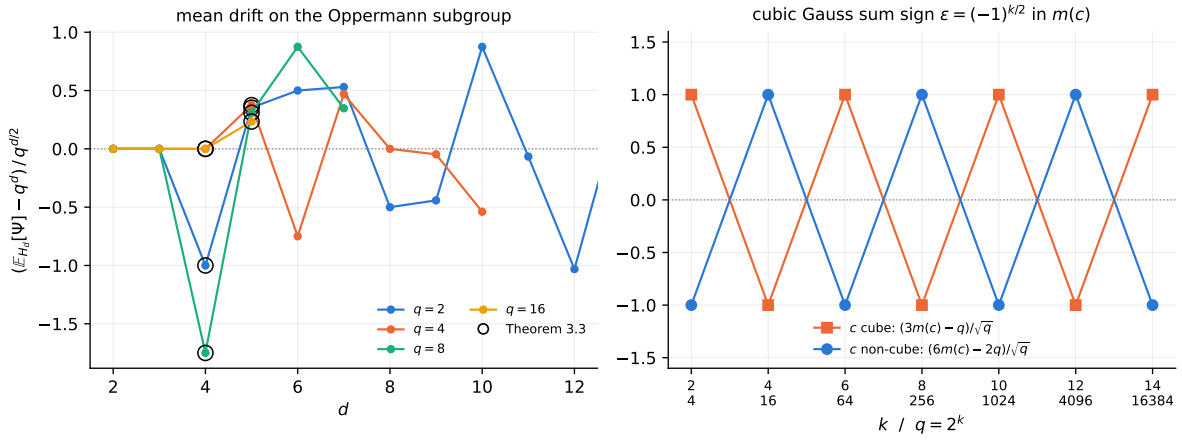



FIGURE 3. *Left:* mean drift on the Oppermann subgroup, $(\mathbb{E}_{H_d}[\Psi] - q^d)/q^{d/2}$, for $q = 2, 4, 8, 16$. Open circles mark the closed forms of Theorem 7.3; the drift vanishes for $d = 2, 3$. *Right:* the normalised values of $m(c)$, which equal $\mp\varepsilon = \mp(-1)^{k/2}$ by Theorem 5.3.

*Coherent superposition.* The $|H_d^\perp| = q^{\lfloor d/2 \rfloor}$ characters in one coset are orthogonal on $G_d$ but coincide on $H_d$, so their coefficients add as amplitudes rather than as energies. If the phases of the $\psi_{2d}(\chi\eta)$ in a coset were independent, $\mathrm{Var}_{H_d}$ and $\mathrm{Var}_{G_d}$ would agree on average. If they are aligned, $\mathrm{Var}_{H_d}$ can exceed $\mathrm{Var}_{G_d}$ by a factor up to $|H_d^\perp|$; if they cancel, it is smaller. Both extremes occur.

**Theorem 7.5.** *Let $q = 2^k$.*

(1) $\mathrm{Var}_{H_2}(\Psi)/q^2 = q - 1$ *and* $\mathrm{Var}_{G_2}(\Psi)/q^2 = 1 - 1/q$. *The ratio is exactly $q = |H_2^\perp|$, so the coherence is maximal.*
(2) $\mathrm{Var}_{H_2}(4\pi)/q^2 = q - 1$.
(3) $\mathrm{Var}_{H_3}(\Psi)/q^3 = 1 - 1/q$ *for $k$ odd and* $3(1 - 1/q)$ *for $k$ even.*

(4) $\mathrm{Var}_{H_3}(6\pi)/q^3 = 0$ *for* $k$ *odd and* $6(1-1/q)$ *for* $k$ *even.*

*Proof.* (1) By Proposition 6.1, $\Psi(x)-q^2$ takes three values: $-(-1)^k(q^2-q)$ on the identity class, $(-1)^k q$ on the other $q-1$ classes of $H_2$, and 0 on the $q^2-q$ classes outside $H_2$. Averaging the squares over $H_2$ (with $q$ classes) and over $G_2$ (with $q^2$ classes) gives $q^2(q-1)$ and $q(q-1)$.

The ratio is maximal because Proposition 6.1(3) gives $\psi_4(\chi) = -(-1)^k q^2$ for all characters of exact level 2: every coset sum is fully aligned.

(2) $E(B_e) = q$ for every $e$ Proposition 6.1(1), so $4\pi = \Psi - q$ on $H_2$ and has the same variance as $\Psi$.

(3), (4) $H_3 \cong \mathbb{F}_q^2$ with coordinates $(e, c)$. By Corollary 5.4, both $\Psi$ and $\pi$ depend only on $c$. Averaging over $(e, c) \in H_3$ is therefore the same as averaging over $c \in \mathbb{F}_q$, and all variances below are written as $q^{-1}\sum_{c\in\mathbb{F}_q}$.

For $k$ odd, $6\pi = q(q^2-1)$ is constant, so $\mathrm{Var}_{H_3}(6\pi) = 0$. The variance of $\Psi$ then comes entirely from the prime powers, $E(C_{e,c}) = q^2[c=0]$. Since $\Psi - q^3 = -q + q^2[c=0]$,

$$\mathrm{Var}_{H_3}(\Psi) = q^{-1}((q^2-q)^2 + (q-1)q^2) = q^2(q-1).$$

For $k$ even, the deviation $\Psi - q^3$ is

- $-q^2 + q$ for $c = 0$;
- $q - 2\varepsilon q^{3/2}$ for each of the $(q-1)/3$ cubes $c$;
- $q + \varepsilon q^{3/2}$ for each of the $2(q-1)/3$ non-cubes $c$.

Its mean is 0. In the second moment the cross terms in $\varepsilon q^{5/2}$ cancel, leaving

$$\mathrm{Var}_{H_3}(\Psi) = q^{-1}(q^2(q-1)^2 + (q-1)(q^2+2q^3)) = 3q^2(q-1).$$

For $6\pi$, the mean is $q(q^2-1)$, and the deviations are

- $-2q(q-1)$ for $c = 0$;
- $2q - 2\varepsilon q^{3/2}$ for each cube $c$;
- $2q + \varepsilon q^{3/2}$ for each non-cube $c$.

The same computation gives

$$\mathrm{Var}_{H_3}(6\pi) = q^{-1}(4q^2(q-1)^2 + (q-1)(4q^2+2q^3)) = 6q^2(q-1).$$

Adding $E(C_{e,0}) = q^2$ halves the deviation at $c = 0$, to $-q(q-1)$. Removing the mean of $E$ shifts the other deviations by $-q$. Together these halve the variance. □

All four statements were also checked exactly, with rational arithmetic on the class counts, for $q = 2, 4, \ldots, 64$ (`verify_variance23.py`).

For $d \geq 4$ we have only numerical data (Table 6, Figure 4). We observe the following:

(1) On the whole group, $\mathrm{Var}_{G_d}(\Psi)/q^d$ follows the Keating–Rudnick value $d-1$ closely for $q \geq 4$, and in odd characteristic. For $q = 2$ it stays near $d-2$. Keating and Rudnick prove $\mathrm{Var} \sim q^{h+1}(n-h-2)$ as $q \to \infty$ for intervals of size $q^{h+1}$ in degree $n$; here $h = d-1$ and $n = 2d$, see [6] for the precise hypotheses. We use $d-1$ only as a reference value.

(2) On the subgroup, $\mathrm{Var}_{H_d}(\Psi)/q^d$ is erratic. For even $d$ it is large and grows with $q$: $q-1$ at $d = 2$, and 1.1, 3.0, 5.0, 18.5 at $d = 4$ for $q = 2, 4, 8, 16$. For odd $d$ it is moderate: $1-1/q$ or $3(1-1/q)$ at $d = 3$, and 1.5, 6.3, 3.5, 6.5 at $d = 5$. This is the coherence of Proposition 7.4.

Two structural features separate even from odd $d$, although we have no proof of the dichotomy beyond $d = 3$. First, the coherence multiplicity $|H_d^\perp| = q^{\lfloor d/2 \rfloor}$ jumps by a factor $q$ at each even $d$. Second, the squares $\varpi^2 = \sum a_i^2 t^{2i}$ of irreducibles of degree $d$ contain only even powers of $t$. For odd $d$ their coefficient $e_d$ (of $t^d$) is 0, so they pile up on the slice $e_d = 0$ of $H_d$. For even $d$ it is $a_{d/2}^2$, which is spread over all $e_d \in \mathbb{F}_q$.

TABLE 6. Normalised variances in characteristic 2: over the Oppermann subgroup $H_d$ and over all classes $G_d$, for $\Psi$ and for $2d\pi$, with the Keating–Rudnick value $d-1$. The entries at $d=2,3$ agree exactly with Theorem 7.5. For $q=2$ only $d\le 12$ is shown.

| $q$ | $d$ | $\lvert H_d\rvert$ | $\frac{\mathrm{Var}_H(\Psi)}{q^d}$ | $\frac{\mathrm{Var}_H(2d\pi)}{q^d}$ | $\frac{\mathrm{Var}_G(\Psi)}{q^d}$ | $\frac{\mathrm{Var}_G(2d\pi)}{q^d}$ | $d-1$ |
|---|---|---|---|---|---|---|---|
| 2 | 2 | 2 | 1.000 | 1.000 | 0.500 | 0.750 | 1 |
| 2 | 3 | 4 | 0.500 | 0.000 | 1.750 | 1.617 | 2 |
| 2 | 4 | 4 | 1.125 | 2.000 | 2.031 | 2.938 | 3 |
| 2 | 5 | 8 | 1.500 | 1.562 | 2.234 | 2.219 | 4 |
| 2 | 6 | 8 | 9.594 | 9.000 | 6.184 | 6.099 | 5 |
| 2 | 7 | 16 | 4.375 | 5.359 | 4.818 | 4.909 | 6 |
| 2 | 8 | 16 | 5.002 | 5.250 | 5.500 | 5.637 | 7 |
| 2 | 9 | 32 | 7.022 | 7.119 | 7.828 | 7.879 | 8 |
| 2 | 10 | 32 | 9.152 | 9.375 | 7.854 | 7.836 | 9 |
| 2 | 11 | 64 | 11.896 | 11.757 | 9.092 | 9.105 | 10 |
| 2 | 12 | 64 | 13.821 | 13.192 | 10.302 | 10.339 | 11 |
| 4 | 2 | 4 | 3.000 | 3.000 | 0.750 | 0.938 | 1 |
| 4 | 3 | 16 | 2.250 | 4.500 | 2.438 | 2.953 | 2 |
| 4 | 4 | 16 | 3.047 | 3.000 | 1.315 | 1.371 | 3 |
| 4 | 5 | 64 | 6.328 | 9.668 | 4.902 | 5.098 | 4 |
| 4 | 6 | 64 | 6.882 | 6.961 | 4.076 | 4.114 | 5 |
| 4 | 7 | 256 | 8.285 | 9.654 | 6.273 | 6.291 | 6 |
| 4 | 8 | 256 | 8.601 | 8.707 | 6.801 | 6.805 | 7 |
| 4 | 9 | 1 024 | 6.827 | 8.703 | 7.733 | 7.742 | 8 |
| 4 | 10 | 1 024 | 11.731 | 11.766 | 8.653 | 8.655 | 9 |
| 8 | 2 | 8 | 7.000 | 7.000 | 0.875 | 0.984 | 1 |
| 8 | 3 | 64 | 0.875 | 0.000 | 2.734 | 2.618 | 2 |
| 8 | 4 | 64 | 5.045 | 4.812 | 3.490 | 3.556 | 3 |
| 8 | 5 | 512 | 3.466 | 2.606 | 3.540 | 3.525 | 4 |
| 8 | 6 | 512 | 19.431 | 19.380 | 4.575 | 4.573 | 5 |
| 8 | 7 | 4 096 | 8.740 | 10.499 | 6.123 | 6.126 | 6 |
| 16 | 2 | 16 | 15.000 | 15.000 | 0.938 | 0.996 | 1 |
| 16 | 3 | 256 | 2.812 | 5.625 | 2.871 | 3.043 | 2 |
| 16 | 4 | 256 | 18.519 | 18.750 | 3.149 | 3.153 | 3 |
| 16 | 5 | 4 096 | 6.478 | 9.532 | 3.991 | 4.003 | 4 |

## 8. QUESTIONS

(1) In characteristic 2, are (A)–(C) the only empty Oppermann sub-intervals for all $q$ and $d$? For $d\le 3$ the question is settled for every $q$ (Theorems 1.1 and 5.2). For each $d\ge 4$, by Corollary 6.2, only finitely many $q$ can carry further empty sub-intervals. Corollary 3.7 shows that, from $d=4$ on, the additive mechanism of (B) is no longer available to produce empty or full classes.

(2) For odd $q$, is every Oppermann sub-interval nonempty, for all $q$ and $d$? Lemma 2.1 turns this into a statement about all short intervals of length $q^{n/2}$. For $d=2$ it holds for every odd $q$ (Theorem 1.4). For each $d\ge 3$, by Corollary 6.2, it fails for at most finitely many $q$. The trace-form method of Section 4 would need, for $d=3$, the joint distribution of $\mathrm{Tr}(x^2)$ and $\mathrm{Tr}(x^3)$ on $\ker \mathrm{Tr}\subset \mathbb{F}_{q^6}$.

(3) Outside the characteristic-2 cases above, is there a constant $c>0$ with $\min_x \pi(x)\ge c\,q^d/2d$ for all large $d$?

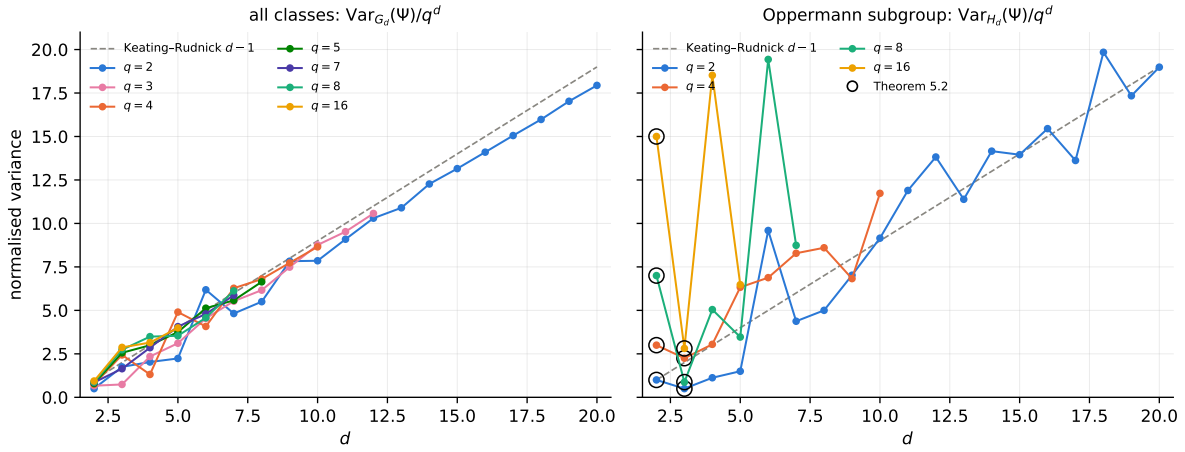



FIGURE 4. Normalised variance of $\Psi$ over all classes $G_d$ (left, odd and even $q$) and over the Oppermann subgroup $H_d$ (right, $q$ even). Dashed line: the Keating–Rudnick value $d-1$. Circles: Theorem 7.5.

(4) In characteristic 2, evaluate the mean drift $\mathbb{E}_{H_d}[\Psi] - q^d$ for $d \geq 6$, and explain the even/odd dichotomy of $\mathrm{Var}_{H_d}(\Psi)/q^d$ observed in Table 6.

## DATA AND CODE

Sections 2–6, except §2.1 and §3.4, are supported by the repository `https://github.com/Ruqing1963/oppermann-function-fields`:

- `hayes_oppermann.py` (Theorems 1.2, 1.3);
- `verify_pointwise_oppermann.py`, `compare_bruteforce.py` (cross-checks);
- `char2_d2_blocks.py`, `char2_d2_theorem.py` (Theorem 1.1(B));
- `swan_parity.py` (Table 1);
- `odd_d2_formula.py` (Theorem 1.4);
- `char2_d3_p3zero.py`, `char2_d3_all.py`, `make_unified_d3_table.py` (Theorem 5.2, Table 2);
- `char2_d2_rigidity.py` (Proposition 6.1);
- `make_figure.py`, `make_figure2.py`, `make_tables.py` (Figures 1, 2; Tables 3, 4).

Section 7, §2.1, §3.4 and Theorem 5.3 are supported by `https://github.com/Ruqing1963/oppermann-char2-subgroup`:

- `verify_subgroup_mean.py` (Theorems 2.4, 2.5, 7.3);
- `cubic_gauss_m.py` (Theorem 5.3);
- `subgroup_variance_census.py`, `verify_variance23.py` (Theorem 7.5, Table 6);
- `verify_additive_slices.py` (Theorem 3.6);
- `make_figures.py` (Figures 3, 4).

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